

Theory-Guided Instance-Dependent Early Stopping via Second-Order Loss Dynamics

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Abstract

Instance-dependent early stopping (IES) speeds up training by skipping the backward pass on individual examples once they look “mastered,” based on how much an example’s loss changes over consecutive epochs. It works well in practice, but its detection threshold and its patience – how long to wait before excluding an example – are hand-tuned constants with no theoretical grounding. We develop TG-IES, a theory-guided version of IES, and close these two gaps to different degrees. First, we show why a second-order loss difference is the right signal in a well-defined noise regime, and use that analysis to derive its detection threshold directly from the training run’s own measured noise rather than hand-tuning it. Second, we derive a conditional, local bound on how much removing one example can perturb the trained model, and use it to compute a theory-motivated patience. We are upfront about where this theory falls short in practice: the derived patience bound is so loose that a fixed configuration ceiling, not the theory, governs every experiment we ran, so only the threshold is genuinely theory-derived; and a direct probe finds that roughly half of excluded instances violate the bound’s own per-instance gradient assumption at any point in training, even though the trained model still ends up close to one trained with no exclusion at all – a tension we report rather than resolve. Empirically, TG-IES matches standard training’s accuracy within about one point while skipping 53–58% of backward passes across several CNN architectures (ResNet-18, VGG-16, DenseNet-121) on CIFAR-10 and CIFAR-100, ahead of three modern pruning methods (InfoBatch, BLS, AlignPrune) and of IES itself at each method’s own stopping point; at a matched compute budget this ordering can reverse at low budgets. The mechanism transfers to a Vision Transformer, with smaller savings and a modest accuracy cost. A controlled ablation shows nearly all of the savings come from the derived threshold rather than the derived patience, and a naive grid search over the threshold reaches a comparable trade-off – so the theory’s main practical benefit is reaching that trade-off automatically, not beating brute-force tuning.

1 Introduction

Training compute for state-of-the-art models has grown enormously (Sevilla et al., 2022), and a training run’s energy use and carbon footprint scale directly with the compute it consumes (Patterson et al., 2022); algorithmic reductions in the number of forward and backward passes a run performs are one of the few practical levers for this cost that do not require more or newer hardware (Bartoldson et al., 2023). Training modern networks means recomputing gradients for every example at every epoch, even for examples classified correctly with high, stable confidence hundreds of epochs before training ends. A growing line of work removes this redundancy, either by pruning the dataset once before training using a static difficulty score (Toneva et al., 2019; Paul et al., 2021; Swayamdipta et al., 2020; Sorscher et al., 2022; Zheng et al., 2023; Maharana et al., 2024), or by dynamically deciding, epoch by epoch, which examples still need a gradient update (Loshchilov & Hutter, 2015; Katharopoulos & Fleuret, 2018; Jiang et al., 2019; Qin et al., 2024; Mindermann et al., 2022; Bankes et al., 2024; Nagaraj et al., 2025; Jin et al., 2026).

Yuan et al. (2025) introduced Instance-dependent Early Stopping (IES), a dynamic method with a simple criterion: stop backpropagating on instance i once

$$|\Delta^2 L_i(\theta^{(t)})| = |L_i^{(t)} - 2L_i^{(t-1)} + L_i^{(t-2)}| < \delta, \quad (1)$$

the magnitude of the backward second-order finite difference of its own loss trajectory, held below δ for a patience window. IES reports strong empirical results, but Equation (1) was motivated empirically: the original paper does not derive why a second-order difference identifies mastery, does not quantify the noise trade-off against first- or third-order differences, does not give a correct curvature reading of what $\Delta^2 L_i$ measures, and does not provide a computable safety guarantee for how long an excluded instance may safely remain excluded. Both δ and the patience are hand-tuned constants.

Research question. Does the IES mastery criterion admit theoretical grounding under explicit, checkable assumptions – grounding that can replace IES’s hand-tuned detection threshold with a quantity derived from the training run itself, and provide a principled, checkable bound on how long an excluded instance may safely stay excluded, while preserving IES’s accuracy–compute trade-off?

Contributions. We make three contributions, each closing a specific gap left open by Yuan et al. (2025).

- Understanding the mastery signal (Sections 4 to 6). We show that a specific ratio of consecutive loss changes cancels an instance’s absolute scale and depends only on its convergence rate (Theorem 4.2); characterize a signal-to-noise band under which comparing exactly two epochs back is MSE-optimal among one, two, or three (Theorem 5.1 and Proposition 5.2); and give an exact decomposition of the second-order difference showing it is not a pure curvature estimate (Theorem 6.1).
- Theory-guided threshold calibration (Section 5). We derive δ directly from the training run’s own measured noise rather than hand-tuning it (Corollary 5.3); a controlled ablation shows this calibration accounts for essentially all of TG-IES’s savings advantage over IES (Section 10.1).
- Conditional stability analysis and empirical evaluation (Sections 7, 7.1 and 8). We derive a local bound on how much excluding one instance can perturb the trained model under an explicit shared-reference idealization (Theorem 7.1), and use it to define a conditional exclusion bound K_{safe} (Corollary 7.3). Under further idealizations we relate the mastery signal to the per-instance gradient norm the bound actually needs (Theorem 7.8 and Corollary 7.9), which we treat as a conditional theoretical bridge rather than a guarantee about the algorithm’s removal rule. We then test these assumptions directly: K_{safe} turns out so large it never binds in any run reported here, so the patience actually used is a fixed ceiling rather than the derived value (Remark 7.5), and a direct probe finds the bound’s own per-instance gradient-floor hypothesis violated by roughly half the excluded set throughout training, even though overall parameter divergence from full-data training still plateaus (Section 10.2 and Appendix T; Appendix E investigates this coexistence quantitatively). Across CIFAR-10/100 and multiple CNN and ViT architectures, TG-IES reduces backward computation while matching baseline accuracy in every CNN setting, and the mechanism transfers to a Vision Transformer at reduced magnitude and a small accuracy cost.

Putting these together gives TG-IES (Algorithm 1): it derives δ from the training run itself and computes a patience seed K^* from the same conditional bound, though in every reported run K^* saturates a fixed ceiling rather than the derived value governing behavior (Remark 7.5). We evaluate against a no-removal baseline, IES exactly as shipped, our literal reimplementations of IES’s stated Algorithm 1, and three modern dynamic-pruning baselines, on a seed-averaged matrix spanning two datasets, four architectures, and two optimizers. TG-IES matches accuracy and gives the largest measured compute saving among the compared methods in every convolutional setting; on a Vision Transformer the advantage transfers at smaller magnitude and a small accuracy cost. Figure 1 summarizes the pipeline.

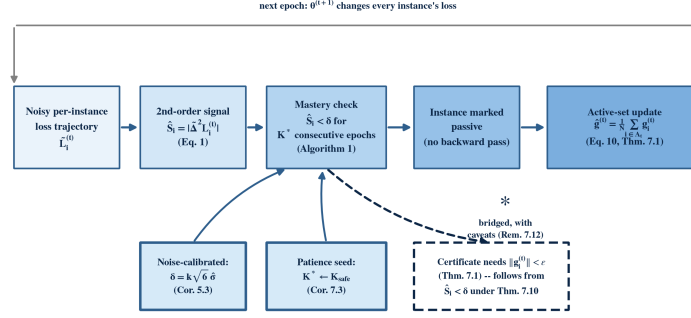


Figure 1: How TG-IES turns a noisy per-instance loss trajectory into a removal decision. δ and K^* are theory-guided (orange); the mastery check they feed (blue) links to the conditional stability bound that justifies removal (purple) via Section 7.1, under the idealizations Remark 7.10 states; Section 10.1 shows what goes wrong when those idealizations do not hold closely enough.

2 Related Work

Static data pruning commits to a fixed subset before or shortly after training starts, using forgetting events (Toneva et al., 2019), early gradient-norm scores (Paul et al., 2021), training-dynamics cartography (Swayamdipta et al., 2020), a cheap proxy model (Coleman et al., 2020), or scaling-law/coverage-diversity arguments (Sorscher et al., 2022; Zheng et al., 2023; Maharana et al., 2024). Dynamic, online selection instead makes a per-epoch decision: importance sampling (Loshchilov & Hutter, 2015; Katharopoulos & Fleuret, 2018), probabilistic backward-pass skipping (Jiang et al., 2019), holdout-model criteria (Mindermann et al., 2022; Bankes et al., 2024), epoch-mean-loss soft pruning (Qin et al., 2024), alignment-score pruning (Qin et al., 2026; Nagaraj et al., 2025), and EMA-smoothed importance (Zhou et al., 2026); Jin et al. (2026) give a different formal guarantee than the conditional bound derived here. None of these – IES included – derive a removal criterion from a stability guarantee (Section 8 and Appendix L compare directly against InfoBatch, BLS, AlignPrune). Curriculum learning (Bengio et al., 2009; Soviany et al., 2022) changes when an example enters, not leaves, training. Appendix A extends this discussion.

3 Preliminaries and Notation

Let $D = \{(x_i, y_i)\}_{i=1}^N$ be the training set and $\theta^{(t)} \in \mathbb{R}^d$ the parameters at epoch t . The per-instance loss is $L_i(\theta) = \ell(f_\theta(x_i), y_i)$, $L_i^{(t)} := L_i(\theta^{(t)})$, and $L(\theta) = \frac{1}{N} \sum_i L_i(\theta)$. Write $g_i^{(t)} := \nabla_\theta L_i(\theta^{(t)})$, $g^{(t)} := \nabla_\theta L(\theta^{(t)}) = \frac{1}{N} \sum_i g_i^{(t)}$, $H_i(\theta) := \nabla_\theta^2 L_i(\theta)$. The observed loss is $\tilde{L}_i^{(t)} := L_i^{(t)} + \xi_{i,t}$ (Assumption 3.3). Finite differences: $\Delta^0 L_i^{(t)} = L_i^{(t)}$, $\Delta^N L_i^{(t)} = \Delta^{N-1} L_i^{(t)} - \Delta^{N-1} L_i^{(t-1)}$. Moments $\mathbb{E}[\cdot]$, $\text{Var}(\cdot)$, $\text{CV}(\cdot) := \sqrt{\text{Var}(\cdot)/\mathbb{E}[\cdot]}$ over instances are w.r.t. $I \sim \text{Unif}\{1, \dots, N\}$. Gradient descent: $\theta^{(t+1)} = \theta^{(t)} - \eta g^{(t)}$.

All results below hold under one or more explicit idealizations, stated once here so each theorem can cite exactly which it needs; all proofs are in Appendix C unless noted, and Appendix B discusses why each idealization is needed and what its violation would do.

Assumption 3.1 (β -smoothness). L is β -smooth: $\|\nabla_\theta L(\theta) - \nabla_\theta L(\theta')\| \leq \beta \|\theta - \theta'\|$ for all θ, θ' .

Assumption 3.2 (Local geometric trajectory). Near an optimum θ^* , the per-instance excess loss follows $L_i^{(t)} - L^* = c_i r_i^t$, $c_i > 0$, $r_i \in (0, 1)$, over the analysis window – a local, near-convergence approximation; Sections 4 and 5 are exact conditional on it.

Assumption 3.3 (Noise: time-independence, finite variance). $\tilde{L}_i^{(t)} = L_i^{(t)} + \xi_{i,t}$ where, conditional on $\theta^{(t)}$, $\xi_{i,t}$ is mean-zero and independent across t for fixed i , with $\text{Var}(\xi_{i,t}) = \sigma^2 < \infty$ (Appendix M tests this directly).

Assumption 3.4 (Sub-Gaussian noise). In addition to Assumption 3.3, each $\xi_{i,t}$ is σ^2 -sub-Gaussian conditional on $\theta^{(t)}$.

Assumption 3.5 (Small rate dispersion). There exist $\bar{r} \in (0, 1)$, $\delta_r \in (0, 1)$ with $r_i \in [\bar{r} - \delta_r, \bar{r} + \delta_r]$ for all i , δ_r small enough for Proposition 4.3’s expansion to hold – an important assumption, not a technicality: Appendix G exhibits widely dispersed rates for which CV moves the wrong way as N increases.

Two further technical conditions – uniform bounds $0 < c_{\min} \leq c_i \leq c_{\max} < \infty$ on Assumption 3.2’s amplitude (Assumption B.1), and a bound M_3 on third derivatives near the trajectory (Assumption B.2) – are stated in full in Appendix B and used by Proposition 4.3 and Theorem 6.1 respectively.

4 Level 1: Scale Cancellation Near Convergence

Lemma 4.1 (Finite differences of geometric sequences). Under Assumption 3.2, $\Delta^N(L_i^{(t)} - L^*) = c_i r_i^{t-N} (r_i - 1)^N$.

Theorem 4.2 (Exact ratio cancellation). Under Assumption 3.2, for any $N \geq 1$ and instance i with $\Delta^{N-1}(L_i^{(t)} - L^*) \neq 0$,

$$\frac{\Delta^N(L_i^{(t)} - L^*)}{\Delta^{N-1}(L_i^{(t)} - L^*)} = \frac{r_i - 1}{r_i}. \quad (2)$$

The ratio depends only on r_i , not c_i ; the denominator is nonzero for all finite t .

Theorem 4.2 gives theorem-level content to IES’s informal “scale invariance” claim: a ratio identity eliminating an instance’s amplitude c_i , not a claim that $\Delta^2 L_i$ itself is scale-free.

Proposition 4.3 (CV closeness under small rate dispersion). Under Assumptions 3.2, 3.5 and B.1, let $X_i(N) := |c_i r_i^{t-N} (1 - r_i)^N|$ for $N \in \{1, 2\}$ at fixed t . There is a finite constant $C(t, \bar{r}, c_{\min}, c_{\max})$ (explicit form in Appendix F) such that, for all sufficiently small δ_r , $|\text{CV}(X(2)) - \text{CV}(X(1))| \leq C(t, \bar{r}, c_{\min}, c_{\max}) \cdot \delta_r$.

5 Level 2: Noise-Optimality of $N = 2$

Theorem 5.1 (Exact bias–variance decomposition). Under Assumptions 3.2 and 3.3,

$$\mathbb{E}[(\Delta^N \tilde{L}_i^{(t)})^2] = c_i^2 r_i^{2(t-N)} (r_i - 1)^{2N} + \binom{2N}{N} \sigma^2. \quad (3)$$

Variance scales combinatorially in N ($\binom{2}{1}=2$, $\binom{4}{2}=6$, $\binom{6}{3}=20$), the quantitative reason high-order differencing is costly.

Proposition 5.2 ($N = 2$ is MSE-optimal for $r_i > \frac{1}{2}$). Under Assumptions 3.2 and 3.3, define $\text{MSE}(N, t) := c_i^2 r_i^{2(t-N)} (r_i - 1)^{2N} + \binom{2N}{N} \sigma^2$ and $\alpha_i := (1 - r_i)^2 / r_i^2$. Then

$$\text{MSE}(2, t) \leq \text{MSE}(1, t) \iff c_i^2 r_i^{2t} \alpha_i (1 - \alpha_i) \geq 4\sigma^2, \quad (4)$$

$$\text{MSE}(2, t) \leq \text{MSE}(3, t) \iff c_i^2 r_i^{2t} \alpha_i^2 (1 - \alpha_i) \leq 14\sigma^2. \quad (5)$$

If $r_i > \frac{1}{2}$ ($\alpha_i < 1$), $N = 2$ minimizes MSE over $\{1, 2, 3\}$ whenever $\frac{4\sigma^2}{\alpha_i(1-\alpha_i)} \leq c_i^2 r_i^{2t} \leq \frac{14\sigma^2}{\alpha_i^2(1-\alpha_i)}$, a band that is non-empty for every $\alpha_i < 1$.

Empirical test (Appendix P). A naive population-average test finds $N=1$ beats $N=2$ – the opposite of Proposition 5.2 – dominated by already-converged instances outside the SNR band. The fair test evaluates each instance at the epoch its own signal first drops below the calibrated $\delta=0.01597$: there, pooling 46,779 instances, $N=2$ gives the lowest MSE, two orders of magnitude below $N=1$ and three below $N=3$, confirming Proposition 5.2 where it applies. Separately, per-quartile rate estimates show mean $\hat{r}$ rising ($0.53 \rightarrow 0.76$) and CV falling ($1.05 \rightarrow 0.75$) across training – Assumption 3.2 fits better in the second half.

Corollary 5.3 (Threshold calibration). Under Assumptions 3.3 and 3.4, $\Delta^2 \xi_{i,t}$ is mean-zero with variance $6\sigma^2$. Setting $\delta = k\sqrt{6}\sigma$: if approximately Gaussian, $\Pr(|\Delta^2 \xi_{i,t}| > \delta) \approx 2(1 - \Phi(k))$; in general, $\Pr(|\Delta^2 \xi_{i,t}| > \delta) \leq 2 \exp(-k^2/2)$.

Corollary 5.3 turns δ from a hand-tuned constant into a function of an estimated noise scale $\hat{\sigma}$ and a target tail probability k , used online by TG-IES (Section 7).

6 Level 3: What the Second-Order Difference Measures

Theorem 6.1 (Backward second-difference decomposition). Under Assumption B.2, along full-batch GD,

$$\begin{aligned} \Delta^2 L_i^{(t)} = & \eta \left(\langle g_i^{(t-2)}, g^{(t-2)} \rangle - \langle g_i^{(t-1)}, g^{(t-1)} \rangle \right) + \frac{\eta^2}{2} g^{(t-1)\top} H_i(\theta^{(t-1)}) g^{(t-1)} \\ & - \frac{\eta^2}{2} g^{(t-2)\top} H_i(\theta^{(t-2)}) g^{(t-2)} + R_{i,t}, \quad |R_{i,t}| \leq \frac{M_3}{6} \eta^3 (\|g^{(t-1)}\|^3 + \|g^{(t-2)}\|^3). \end{aligned} \quad (6)$$

$\Delta^2 L_i$ is not a pure curvature estimate. Equation (6) shows it also contains an $O(\eta)$ gradient-drift term alongside the $O(\eta^2)$ curvature term. In the small-step-size limit $\eta \rightarrow 0$, the drift term is the dominant one; the two terms vanish together only near convergence. This ordering describes the limit $\eta \rightarrow 0$, not this paper’s actual step sizes (η up to 0.1). Direct instrumentation (Appendix K) finds the opposite in absolute magnitude under every schedule tested: the drift term correlates only weakly with the observed signal ($r=0.23\text{--}0.37$) and does not exceed the curvature-plus-remainder term. A reader should therefore not take “drift leads” as a description of what the mastery signal measures in practice – only of the leading order as $\eta \rightarrow 0$. The decomposition itself extends unchanged to Algorithm 1’s active-set gradient (Equation (7)).

7 Removal Perturbation Bounds and a Conditional Exclusion Bound

To keep the update on the same $1/N$ scale as full-batch GD while skipping passive instances, TG-IES uses the scaled active-set gradient

$$\hat{g}^{(t)} = \frac{1}{N} \sum_{i \in A_t} g_i^{(t)}, \quad (7)$$

where A_t is the active set at epoch t – unlike shipped IES’s $1/|A_t|$ normalization, which increases effective step size as instances are excluded; Equation (7) shrinks proportionally, trading that implicit speed heuristic for the stability bound below.

Theorem 7.1 (Conditional local stability bound, shared-reference trajectory). Suppose instance i satisfies $\|g_i^{(t)}\| < \varepsilon$ for all $t \in [T - K, T]$. Consider two runs sharing the same reference trajectory $\{\theta^{(t)}\}_{t=T-K}^T$, both using Equation (7) evaluated on it, differing only in whether $i \in A_t$. Then $\|\theta_{\text{full}}^{(T)} - \theta_{\text{rem}}^{(T)}\| \leq \frac{K\eta\varepsilon}{N}$. If additionally L_{test} is β_{test} -Lipschitz, $|L_{\text{test}}(\theta_{\text{full}}^{(T)}) - L_{\text{test}}(\theta_{\text{rem}}^{(T)})| \leq \beta_{\text{test}} K\eta\varepsilon/N$.

Interpretation. If one excluded instance’s gradient norm stays below ε throughout a K -step window, removing it changes the parameters by at most $K\eta\varepsilon/N$, relative to a run that shares the same reference trajectory. This is a local, single-removal statement: it bounds one counterfactual against a common reference, not the accuracy of the actual trajectory produced by removing many instances over many epochs – the gap the next remark names explicitly.

Remark 7.2 (Why this is conditional and local). Theorem 7.1 compares two runs on the same reference trajectory; an actual run’s two counterfactual trajectories diverge from the moment i is removed. This is conditional (on the shared-reference idealization) and local (one removal’s instantaneous perturbation), not a guarantee that Algorithm 1’s actual, multi-removal trajectory stays bounded from the full-data one over an entire run – nothing here rules out compounding drift. Appendix D attempts a multi-step bound (correct but exponentially loose, Proposition D.1); Section 10.2 instruments the actual quantity directly and finds no blow-up over 200 epochs – evidence against the worst case, not a proof.

Corollary 7.3 (Conditional exclusion bound). Given tolerance $\tau > 0$ on test-loss deviation, any exclusion window $K \leq K_{\text{safe}} := \left\lfloor \frac{N\tau}{\beta_{\text{test}}\eta\varepsilon} \right\rfloor$ keeps $|L_{\text{test}}(\theta_{\text{full}}^{(T)}) - L_{\text{test}}(\theta_{\text{rem}}^{(T)})| \leq \tau$; the floor (not ceiling) is required since rounding up would violate the bound.

Remark 7.4 (K_{safe} is not Algorithm 1’s patience). K_{safe} bounds a closed exclusion window counted backward from a fixed endpoint; the algorithm’s patience K^* governs an open-ended exclusion starting once K^* consecutive low-signal epochs elapse. No result here shows seeding $K^* \leftarrow K_{\text{safe}}$ keeps that exclusion inside Corollary 7.3’s budget; K_{safe} is a principled seed, not a safety proof. Independently, Theorem 7.1 requires $\|g_i^{(t)}\| < \varepsilon$ while the algorithm thresholds the noisy $\hat{S}_i < \delta$; Theorem 6.1’s route (Cauchy–Schwarz on drift) does not close this, since drift is a difference of inner products, not either absolutely – Section 7.1 closes it differently.

Remark 7.5 (K_{safe} is never the binding constraint in this paper’s experiments). Every run clamps K^* to $[1, 15]$, and in every setting tested, including every ViT-Tiny cell (Appendix Q), the uncapped K_{safe} already exceeds 15 before the clamp – so the patience used is the engineering ceiling, not K_{safe} (Table 1: 5 orders of magnitude larger on the primary cell). This limits the claim that TG-IES replaces IES’s hand-tuned patience with a derived quantity: δ is genuinely derived and does the paper’s practical work (Section 10.1), but the patience used is, in every run, the fixed ceiling 15 – as hand-set as IES’s own. Corollary 7.3 demonstrates only that K_{safe} is an upper bound wide enough never to bind at this τ , not a setting where the derived value governs behavior; a tighter τ would test that, and we have not run it.

Table 1: Uncapped K_{safe} vs. the configured cap, primary cell, epoch-30 recalibration ($N=50000$, $\tau=0.02$, $\beta_{\text{test}}=1$, $\eta_{31}\approx 0.0282$).

ε_0 (per seed)	K_{safe} (uncapped, per seed)	Used K^*
0.0269 / 0.0297 / 0.0293	1.32M / 1.19M / 1.21M	15 (all seeds)

7.1 Closing the Gap: From Mastery Signal to Gradient Norm

Assumption 7.6 (Per-instance smoothness). Each L_i is Λ_i -smooth on the same neighborhood U that Assumption 3.2 is stated on, and $L_i(\theta^*) = L^*$ at the same θ^* for every i (a shared interpolating solution).

Lemma 7.7 (Smoothness bounds the gradient by suboptimality). If f is Λ -smooth on U and $\theta^* \in U$ satisfies $f(\theta^*) \leq f(\theta - \frac{1}{\Lambda} \nabla f(\theta))$ for $\theta \in U$, then $\|\nabla f(\theta)\|^2 \leq 2\Lambda(f(\theta) - f(\theta^*))$.

Theorem 7.8 (Mastery bounds the gradient norm). Under Assumptions 3.2 and 7.6 and Lemma 7.7’s hypothesis for $f = L_i$,

$$\|g_i^{(t)}\| \leq \frac{r_i}{1-r_i} \sqrt{2\Lambda_i \Delta^2 L_i^{(t)}}, \quad \text{so} \quad \Delta^2 L_i^{(t)} < \delta \Rightarrow \|g_i^{(t)}\| < \varepsilon_i(\delta) := \frac{r_i}{1-r_i} \sqrt{2\Lambda_i \delta}. \quad (8)$$

Corollary 7.9 (Probabilistic bridge under observation noise). Under Theorem 7.8’s assumptions plus Assumption 3.4, for $k > 0$, $\delta' := \delta + k\sqrt{6} \sigma$: $\Pr(\hat{S}_i < \delta \text{ and } \|g_i^{(t)}\| \geq \varepsilon_i(\delta')) \leq 2\exp(-k^2/2)$.

Remark 7.10 (Scope). This closes Remark 7.4’s gap under two further idealizations (Assumption 7.6, a shared interpolating solution), with three caveats: $\varepsilon_i(\delta) \propto r_i/(1-r_i)$ grows unboundedly as $r_i \rightarrow 1$ (uninformative near-stationary rather than near-converged); Corollary 7.9 bounds a joint, not conditional, probability; and, more basically, every experiment here trains with weight decay (5×10^{-4}), which pulls parameters away from any per-instance loss-minimizing θ^* (cross-entropy typically has no finite minimizer under it at all) – a real tension with Assumption 7.6’s literal reading that we do not resolve here. Given these idealizations, we treat Theorem 7.8 and Corollary 7.9 as a conditional theoretical bridge, not as an empirical guarantee that every instance TG-IES excludes has a small gradient – Appendix T’s direct probe checks this empirically and finds the gap the idealizations predict.

Empirical check (Appendix P): since r_i, Λ_i are not estimated per instance, we test Theorem 7.8’s qualitative content instead – confirmed across all three LR schedules, with mastery-passing instances having 30–33× smaller mean gradient norms.

7.2 The TG-IES Algorithm

Algorithm 1 Theory-Guided Instance-Dependent Early Stopping (TG-IES)

Require: data D , model f_θ , learning rate η , noise estimate $\hat{\sigma}$, target per-check false-positive rate $p \in (0, 1)$, tolerance τ , patience ceiling $K_{\max}$, epochs T
 Set $k \leftarrow \Phi^{-1}(1 - p/2)$, $\delta \leftarrow k\sqrt{6}\hat{\sigma}$ {Corollary 5.3}
 Estimate $\varepsilon_0 \leftarrow$ recent per-instance gradient-norm floor
 Set $K_{\text{safe}} \leftarrow \lfloor N\tau/(\beta_{\text{test}}\eta\varepsilon_0) \rfloor$ {Corollary 7.3}
 Set $K^* \leftarrow \min(K_{\text{safe}}, K_{\max})$ {seeds patience per Remark 7.4; K_{safe} never binds in this paper’s runs (Remark 7.5), so $K^* = K_{\max}$ throughout}
 Init $\theta^{(0)}$; $\tilde{L}_i^{(-1)} = \tilde{L}_i^{(-2)} = \infty$; $\text{cnt}[i] = 0$; all i active
 for $t = 1$ to T do
 Forward pass on all $i \in D$: compute $\tilde{L}_i^{(t)}$
 for each $i \in D$ do
 $\hat{S}_i \leftarrow |\tilde{L}_i^{(t)} - 2\tilde{L}_i^{(t-1)} + \tilde{L}_i^{(t-2)}|$
 if $\hat{S}_i < \delta$ then
 $\text{cnt}[i] += 1$
 else
 $\text{cnt}[i] \leftarrow 0$; mark i active
 end if
 if $\text{cnt}[i] \geq K^*$ then
 mark i passive
 end if
 end for
 $A_t \leftarrow \{i : i \text{ active}\}$; break if $A_t = \emptyset$ or stopping criterion triggered
 Backward pass on A_t : $\hat{g}^{(t)} \leftarrow \frac{1}{N} \sum_{i \in A_t} \nabla_{\theta} L_i(\theta^{(t)})$ {Equation (7)}
 $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \hat{g}^{(t)}$
end for
return $\theta^{(T)}$

Practical notes (Appendix H): one-time recalibration at epoch 30; ε_0 from a cheap last-layer proxy; consistent forward-pass discipline avoiding a train/eval batch-norm mismatch (Ioffe & Szegedy, 2015).

7.3 Scope of Theory-Guided Calibration

TG-IES derives δ from measured quantities ($\hat{\sigma}$) rather than hand-tuning it, and this is where the paper’s practical work happens (Section 10.1). Patience is weaker than that: K^* is computed via Corollary 7.3, but Remark 7.5 shows the computed K_{safe} never binds at this τ , so the patience used is the fixed ceiling 15 – itself hand-set. This costs four constants fixed once, identical across every setting ($p_{\text{budget}} = 0.3$, $\tau = 0.02$, $\beta_{\text{test}} = 1$, percentile 20; Appendix N sweeps the percentile). Appendix I checks the “same four constants everywhere” claim on all eleven cells TG-IES runs on – true of δ , not, in this sense, of K^* .

8 Experimental Setup

Methods. (1) Baseline: no removal. (2) IES (shipped): released code. (3) IES (literal): our faithful reimplement of Yuan et al. (2025)’s stated Algorithm 1. (4) TG-IES: Algorithm 1. All four share byte-identical data pipelines, model code, and optimizer configuration.

Data, models, optimization. CIFAR-10/100 (Krizhevsky, 2009); ResNet-18 (He et al., 2016) (primary), VGG-16 (Simonyan & Zisserman, 2015), DenseNet-121 (Huang et al., 2017), and a CIFAR-native ViT-Tiny (Appendix Q). SGD, momentum 0.9, weight decay 5×10^{-4} , batch size 64, 200 epochs; exponential ($\eta_0=0.1$, $\gamma=0.96/\text{epoch}$) and fixed ($\eta=10^{-3}$) schedules, plus Adam (Appendix J). TG-IES: $p_{\text{budget}} = 0.3$, $\tau = 0.02$, $\beta_{\text{test}} = 1$, percentile 20, one-time recalibration at epoch 30, $K^* = \min(K_{\text{safe}}, K_{\max})$ with $K_{\max} = 15$ (Remark 7.5: K_{safe} never binds, so $K^* = K_{\max} = 15$ in every reported run).

Matrix and statistics. All cells except three LR-sweep cells (Appendix K, single-seed by design) run at seeds $\{0, 1, 2\}$ (mean $\pm$ std; Appendix W gives paired statistics). Accuracy is best-over-200-epochs (Yuan et al., 2025); backprop savings are $100 \times (1 -$

Table 2: Which result rests on which idealization, and its practical status in this paper’s experiments.

Result	Main assumption	What it establishes	Practical status
Theorem 4.2	Assumption 3.2 (geometric loss decay)	Scale cancellation	Exact under the model
Proposition 5.2	Assumptions 3.2 and 3.3 + rate dispersion	When $N=2$ minimizes MSE	Conditional on an SNR band; confirmed in-band
Theorem 6.1	Assumption B.2 (bounded third derivatives)	Exact drift/curvature decomposition	Ordering is asymptotic ($\eta \rightarrow 0$); not confirmed in magnitude at this paper’s step sizes
Theorem 7.1	Small per-instance gradient + shared-reference trajectory	Local, single-removal exclusion bound	Conditional; per-instance hypothesis violated for $\sim 47\%$ of the excluded set
Theorem 7.8	Assumption 7.6 + shared interpolating solution	Mastery $\Rightarrow$ small gradient	Highly conditional; in tension with the weight decay used here

Table 3: Primary setting, mean $\pm$ std, 3 seeds, incl. three modern baselines (Appendix L). **Green**: best in column; **yellow**: not best (TG-IES trades a small accuracy cost for a large compute saving, the point of the method).

Method	Test Acc. (%)	Backprop Saved (%)
Baseline	95.03 $\pm$ 0.13	0.03 $\pm$ 0.00
IES (shipped)	94.71 $\pm$ 0.09	45.66 $\pm$ 0.38
IES (literal)	94.85 $\pm$ 0.17	49.02 $\pm$ 0.44
InfoBatch (Qin et al., 2024)	94.32 $\pm$ 0.12	38.91 $\pm$ 0.01
BLS-InfoBatch (Zhou et al., 2026)	94.68 $\pm$ 0.18	35.84 $\pm$ 0.09
AlignPrune (Qin et al., 2026)	94.48 $\pm$ 0.05	20.83 $\pm$ 0.07
TG-IES (ours)	94.85 $\pm$ 0.28	53.63 $\pm$ 0.30

cumulative backprop instances/ $(N \times \text{epochs})$), FLOP-normalized in Section 9.1. All runs used a single RTX 5090 GPU.

9 Results

Appendix W reports paired per-seed differences, CIs, effect sizes, and an exact sign test for every 3-seed setting, including this one; the $\pm$ above is a sample standard deviation, not a significance test.

On all three CNN settings TG-IES sits at or above every other method’s accuracy while saving the most compute among the compared methods (Appendix P plots the trade-off). A reproducibility correction. An earlier version reported TG-IES collapsing to 1.8% savings on VGG-16, traced to a stale, repeated-recalibration configuration (audit trail in Appendix V). Re-run, VGG-16 leads every method (54.4% saved); ResNet-18(fixed-LR) genuinely trails IES (literal) at 29.3% $\pm$ 0.3 vs. 49.4% $\pm$ 0.2 (Table 4), attributed to its fixed, non-decaying learning rate (Appendix K). The underlying failure mode remains real, demonstrated by Section 10.1’s controlled ablation rather than this since-corrected data point.

9.1 FLOP-Normalized Savings, and Measured Efficiency

Every IES-family method forwards every instance every epoch, so backprop-instance savings overstate FLOP savings; the real backward/forward ratio is 1.995–1.998 here, giving FLOP saved = $\frac{2}{3} \times (\text{backprop saved})$, so TG-IES’s 53.63% becomes **35.7%** FLOP-normalized – still largest of the four (30.4%/32.7% IES shipped/literal), our recommended figure for hardware/energy comparison. Measured wall-clock/energy (Appendix O) confirm the ordering, though TG-IES unexpectedly has the highest peak GPU memory (its recalibration

Table 4: Secondary settings, mean $\pm$ std, 3 seeds. R18(F): ResNet-18, fixed LR. VGG-16(E): VGG-16, exponential LR.

Method	R18(F)		VGG-16(E)	
	Acc.	Saved	Acc.	Saved
Baseline	92.21 $\pm$ 0.08	0.03 $\pm$ 0.00	93.30 $\pm$ 0.12	0.03 $\pm$ 0.00
IES (shipped)	91.69 $\pm$ 0.09	44.50 $\pm$ 0.24	93.31 $\pm$ 0.04	37.71 $\pm$ 0.87
IES (literal)	91.77 $\pm$ 0.11	49.42 $\pm$ 0.17	93.15 $\pm$ 0.17	41.02 $\pm$ 1.12
TG-IES	91.91 $\pm$ 0.18	29.27 $\pm$ 0.30	93.11 $\pm$ 0.14	54.43 $\pm$ 0.12

pass), a trade-off we report rather than omit; a batch-size sweep (Appendix O) finds this wall-clock advantage is a full-run property that does not hold in short 8-epoch windows, where TG-IES is slower than baseline at every batch size tested.

10 Generalization, Baselines, and Ablations

A second dataset (CIFAR-100) and a third architecture (DenseNet-121) show the same pattern, more pronounced on CIFAR-100: TG-IES matches baseline accuracy (0.87/0.08-point cost) with the largest saving of all four – 57.76 $\pm$ 0.04% and 54.53 $\pm$ 0.66% respectively (full tables: Appendices I and V).

Beyond convolutions: ViT-Tiny (Appendix Q: full tables, caveats). TG-IES saves 25.52 $\pm$ 0.26% (CIFAR-10) and 26.28 $\pm$ 0.14% (CIFAR-100) – 1.5–2.1 $\times$ IES (literal) – via the same four constants, transferring the mechanism but not its magnitude: much smaller savings than CNNs, with a small accuracy cost not seen there.

Modern dynamic-pruning baselines (Table 3; full comparison Appendix L). TG-IES leads InfoBatch (Qin et al., 2024), BLS-InfoBatch (Zhou et al., 2026), and AlignPrune (Qin et al., 2026) by a wide margin at each method’s own stopping point, though not at a matched compute budget: InfoBatch/BLS-InfoBatch lead by 2.5–4.3 points at 20–30% budget, narrowing and reversing by 40% (Appendix L).

10.1 Component Ablation: Which Mechanism Drives the Gains?

TG-IES differs from IES (literal) in two calibrated quantities: δ (Corollary 5.3) and K^* (Corollary 7.3). Table 5 holds one fixed at IES (literal)’s value while calibrating the other (primary setting, seed 0).

Table 5: Component ablation, primary setting, seed 0. Starred = calibrated; unstarred = held at IES (literal)’s values. Deliberately-crippled TG-IES variants, not competing methods.

Setting	Acc. (%)	Saved (%)	δ	K^*
Neither	94.92	48.59	0.001	1
δ only	94.68	66.62	0.0077*	1
K^* only	95.04	3.41	0.001	15*
Both (full TG-IES)	94.61	53.48	0.0160*	15*

Two findings, neither expected going in. Calibrating δ alone ($K^*=1$) saves more than full TG-IES (66.6% vs. 53.5%) at comparable accuracy: essentially all of TG-IES’s savings advantage traces to threshold, not patience, calibration. Calibrating K^* alone against a fixed strict $\delta=0.001$ collapses savings to 3.4%: the patience counter keeps resetting before accumulating 15 consecutive epochs under strict δ – a controlled demonstration of Remark 7.4’s gap (the VGG-16 anomaly Section 9 corrected had only suggested this uncontrollably): δ, K^* calibrated independently can each be correct w.r.t. its own corollary while jointly failing. Full TG-IES lands between the extremes (53.5%), trading some of δ -only’s raw savings for the additional patience mechanism Theorem 7.1’s stability analysis motivates. Because K_{safe} does not bind in our experiments (Remark 7.5), that patience is in practice the fixed ceiling $K_{\text{max}} = 15$, not the derived bound.

Does calibration beat a naive search? A 5-point grid over fixed δ (Appendix S) traces the same frontier as the calibrated $\delta=0.0077$ (94.68%/66.62%; e.g. $\delta=0.015$: 94.52%/70.47%). This does not show the derived δ beats search – only that calibration reaches a comparable point in one run, not the grid’s five. Normalization ($1/N$ vs. $1/|A_t|$) is not a fifth ablatable axis here, for a concrete, checked reason given in Appendix H.

10.2 K^* -Sweep and Trajectory-Divergence Instrumentation

To probe Remark 7.2’s open question directly, we sweep patience $K^* \in \{1, 2, 4, 8, 15, 30\}$ (self-calibrated δ , primary cell, 3 seeds) and $\log \|\theta_{\text{TG-IES}}^{(t)} - \theta_{\text{baseline}}^{(t)}\|$ against a shared-seed baseline every 20 epochs (matched baseline: $95.08 \pm 0.14\%$).

The accuracy/savings trade-off is smooth and monotonic in K^* – savings fall from $66.98 \pm 0.62\%$ ($K^*=1$) to $44.13 \pm 0.40\%$ ($K^*=30$) as accuracy rises to $94.87 \pm 0.09\%$ (Appendix R), not a cliff. Absolute parameter distance from baseline decreases over training for every K^* and plateaus late, ordered exactly as Theorem 7.1 predicts, no crossings. This does not mean the per-instance hypothesis holds: a reactivation probe finds 47.4% of excluded instances exceed the calibrated floor at every checkpoint, no improving trend (Appendix T) – an unresolved coexistence, not a vindication. Evidence consistent with, not a substitute for, the multi-step bound Appendix D attempts.

11 Conclusion

We asked whether IES’s mastery criterion admits theoretical grounding that replaces its hand-tuned threshold with a training-derived quantity and its hand-tuned patience with a principled, checkable bound. We established five results under explicit idealizations: exact scale-cancellation (Theorem 4.2); a bias-variance/SNR regime for second-order differencing (Theorem 5.1 and Proposition 5.2); a corrected decomposition whose drift-dominance ordering is not confirmed in magnitude at real step sizes (Theorem 6.1, Appendix K); a conditional, local stability bound yielding closed-form δ, K^* (Theorem 7.1), instantiated as TG-IES; and a gradient-norm bridge under further idealizations (Theorem 7.8 and Corollary 7.9). The honest empirical picture is more qualified than a first read of Section 7 suggests: δ is genuinely derived and carries essentially all of TG-IES’s savings advantage; K^* is not, since K_{safe} never binds and the ceiling used is fixed (Remark 7.5), and the bound’s own per-instance hypothesis is violated by roughly half the excluded set at every snapshot when checked directly (Appendix T). The contribution this paper supports is a theory-guided understanding and calibration of IES’s mastery signal, not a complete safety proof for dynamic data removal.

12 Limitations

Theoretical assumptions. Our analysis rests on a local geometric loss-decay model (Assumption 3.2), bounded and time-independent observation noise (Assumption 3.3), and smoothness (Assumptions 3.1 and B.2); Theorem 7.8’s gradient-norm bridge additionally needs per-instance smoothness and a shared interpolating solution (Assumption 7.6). These are useful analytical idealizations, not properties we verify hold throughout neural network training.

Patience calibration. K_{safe} never binds in any setting we test (Remark 7.5): the patience actually used is the fixed engineering ceiling $K_{\text{max}} = 15$, as arbitrary as IES’s own constant. Only the threshold δ is genuinely derived from the training run, and it accounts for essentially all of TG-IES’s savings advantage over IES (Section 10.1); a naive grid search over δ reaches a comparable accuracy-savings frontier, so calibration’s practical advantage is reaching that frontier automatically, not beating brute-force search (Appendix S).

Trajectory-level guarantees. Theorem 7.1 bounds one instance’s removal on a shared reference trajectory, not the accuracy of Algorithm 1’s actual, multi-removal trajectory over an entire run. A first multi-step extension is exponentially loose (Appendix D), and a direct probe finds the bound’s own per-instance gradient-floor hypothesis violated by roughly half the excluded set at every checkpoint (Appendix T), even though measured parameter divergence from full-data training plateaus rather than growing (Section 10.2; Appendix E narrows, but does not close, this gap). We treat the plateau as empirical evidence against the worst case, not as validation of the theoretical bound.

The gradient-norm bridge. Theorem 7.8 and Corollary 7.9 require a shared interpolating solution across instances, in tension with the weight decay used in every experiment here

(Remark 7.10). We treat this result as a conditional theoretical bridge, not a guarantee that every excluded instance has a small gradient.

Empirical scope. All experiments are at CIFAR scale with a limited set of CNN and ViT architectures and optimizers; we do not claim the measured savings transfer directly to larger-scale training. The reported wall-clock and energy advantage is also a full-training-run property: in short 8-epoch windows TG-IES is slower than baseline at every batch size tested (Appendix O), because its per-instance bookkeeping and one-time recalibration pass are not yet amortized. TG-IES should therefore not be read as a universally faster per-epoch training procedure.

Impact Statement

This paper advances compute-efficient training; we identify no notable societal risks.

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A Extended Related Work

Why training-time compute matters. Training compute for state-of-the-art models has grown roughly $10\times$ every year since 2012 and continues to grow (Sevilla et al., 2022), with a directly proportional energy and carbon cost (Patterson et al., 2022). Bartoldson et al. (2023)’s survey of algorithmic (rather than hardware) efficiency gains frames exactly the category of method this paper contributes to: a technique reducing the number of forward/backward passes a fixed training recipe performs, orthogonal to and stackable with hardware- or precision-level efficiency gains. Section 9.1’s FLOP-normalized accounting of TG-IES’s savings is this paper’s own direct, measured contribution to that question.

Static data pruning and difficulty scoring. Toneva et al. (2019) track how often an example is learned then forgotten across epochs and show unforgettable examples can be removed with little accuracy loss. Paul et al. (2021) score examples early in training with the gradient-norm-based EL2N score and prune a static subset before the rest of training. Swayamdipta et al. (2020) map examples along confidence and variability axes computed from training dynamics (“dataset cartography”). Coleman et al. (2020) select a subset using scores from a small, cheap proxy model rather than the target model itself. Sorscher et al. (2022) show, empirically and via a statistical-mechanics analysis of a toy model, that a good pruning metric can convert the usual power-law scaling of test error with dataset size into exponential scaling; Zheng et al. (2023) (CCS) show that pruning to the highest-difficulty examples alone hurts distributional coverage at high pruning rates and instead balance importance against coverage explicitly; Maharana et al. (2024) (D2 Pruning) use a message-passing formulation over a similarity graph to balance the same trade-off. All of these commit to a fixed subset before or shortly after training starts; IES and TG-IES instead make a per-instance, per-epoch decision using the current trajectory.

Dynamic and online example selection. Loshchilov & Hutter (2015) rank minibatches online by loss to bias sampling toward harder examples. Katharopoulos & Fleuret (2018) use importance sampling to reduce gradient variance, building on Needell et al. (2014). Jiang et al. (2019) (Selective-Backprop) skip backpropagation for examples with low relative loss via a probabilistic accept/reject rule, close in spirit to IES but formalized on a different argument (a probabilistic threshold on the loss distribution, versus a finite-difference criterion on the loss trajectory), with no theoretical treatment on either side. Mindermann et al. (2022) (RHO-LOSS) select points simultaneously learnable, worth learning, and not yet learnt via a holdout-model criterion; Bankes et al. (2024) (REDUCR) extend this with class-priority reweighting for robustness, orthogonal to the calibration question here. Qin et al. (2024) (InfoBatch) prune examples below the epoch-mean loss with probability r , rescaling kept low-score instances’ gradients by $1/(1-r)$, annealing back to the full dataset for the final epochs; unlike IES/TG-IES it skips both forward and backward on a pruned example. Qin et al. (2026) (AlignPrune) replace loss-magnitude ranking with a Dynamic Alignment Score (correlation between an example’s loss trajectory and a reference trajectory) for label-noise robustness, a different objective than mastery-based removal; Nagaraj et al. (2025) use a related trajectory-correlation signal for one-shot coreset selection against a held-out reference, with convergence guarantees for the resulting subset – the closest published criterion to IES/TG-IES’s own signal, differing in using a held-out reference rather than a self-referential finite difference, and selecting once rather than every epoch. Zhou et al. (2026) (BLS) use an EMA of loss as a cheap, signal-processing alternative to TG-IES’s noise-calibrated δ for the same denoising problem. Concurrently, Jin et al. (2026) (OrderDP) give a formal guarantee for dynamic pruning via an unbiased ranking surrogate with a lossless-convergence proof, a different mechanism than the per-instance stability bound here – two different answers to the same open question rather than competitors on the same construction. Zhu et al. (2026) address an analogous unbiased-pruning problem for group-based RL policy optimization (GRPO rollout pruning for LLM reasoning), noted for completeness; its setting (sequence-level RL) is different enough that we do not attempt a direct comparison. The same question – what to spend a gradient update on – is asked one level up, over pretraining-data domains, in LLM pretraining (Xie et al., 2023), a parallel literature at a coarser granularity. Wen et al. (2025) (GradES) stops updating individual transformer projection matrices once their gradient norm falls below a threshold, a per-parameter-group decision complementary to,

rather than competing with, TG-IES’s per-instance decision (Appendix Q); we are not aware of a direct empirical comparison in either direction.

Curriculum learning. Bengio et al. (2009) order examples easy-to-hard over training; Soviany et al. (2022) survey the resulting literature. This is complementary to IES/TG-IES: curriculum learning changes when an example enters training, while IES/TG-IES changes when it leaves.

Direct predecessor. Yuan et al. (2025) is the paper this work theoretically grounds; Section 1 details the five gaps closed.

B Extended Discussion of Assumptions

Assumption B.1 (Uniform bounds on c_i). There exist $0 < c_{\min} \leq c_{\max} < \infty$ with $c_i \in [c_{\min}, c_{\max}]$ for all i (Assumption 3.2’s amplitude).

Assumption B.2 (Bounded third derivatives). There exists $M_3 < \infty$ and a neighborhood U of the GD trajectory such that for all i , all $\theta \in U$, and unit vectors u, v, w : $|\nabla^3 L_i(\theta)[u, v, w]| \leq M_3$.

Remark B.3 (Why β -smoothness, and where it is used). Assumption 3.1 caps how fast the gradient can change as parameters move; it is the standard lever turning a single-step bound into one holding over many steps. Appendix D uses it directly for exactly this purpose: a first attempt extending Theorem 7.1’s single-step bound to a multi-step bound. That attempt succeeds only partially – a genuine but exponentially-loose bound, not a practically useful one over hundreds of epochs.

Remark B.4 (Scope of the geometric-trajectory assumption). Assumption 3.2 is a local, near-convergence approximation, not a global description of training: loss trajectories in practice are non-monotonic and noisy, and r_i varies over time. Section 5’s quartile analysis finds it a better approximation in the second half of training than the first.

Remark B.5 (Noise-model realism). Assumption 3.3 is an idealization: SGD revisits overlapping minibatches, correlating noise across nearby t ; Assumption 3.4 may also be violated, since deep-learning gradient noise can be heavy-tailed. Under correlated or heavy-tailed noise, Theorem 5.1’s variance formula becomes a lower bound and Corollary 5.3’s tail bound may be optimistic. Appendix M tests time-independence directly and finds the measured autocorrelation small and consistent with a detrending-method artifact rather than real correlation, correcting Theorem 5.1 at $N=2$ by under 1.3% even under the conservative reading.

Remark B.6 (Why small rate dispersion). r_i is instance i ’s per-epoch decay rate; Assumption 3.5 says different instances’ rates do not spread too far from a common $\bar{r}$, letting Proposition 4.3 treat $\text{CV}(X(2))$ as a small perturbation of $\text{CV}(X(1))$ via a first-order Taylor argument valid only when the spread is genuinely small. Appendix G shows explicitly what breaks when it is not.

Remark B.7 (Why bounded c_i). c_i is instance i ’s excess-loss amplitude; bounding it away from 0 and ∞ keeps $\text{CV}(\cdot)$ well-defined and the perturbation constant in Proposition 4.3 finite, rather than dominated by one pathological instance.

Remark B.8 (Why bounded third derivatives). This bounds how quickly curvature itself can change, keeping Theorem 6.1’s Taylor remainder genuinely smaller than the $O(\eta^2)$ curvature term it isolates; without it, “curvature is $O(\eta^2)$, drift is $O(\eta)$ ” would not be a controlled statement.

Remark B.9 (An unused assumption, kept for future work). A bounded gradient envelope on a window ($G := \sup_t \|g^{(t)}\| < \infty$) is a natural companion to β -smoothness for a future contraction argument bounding trajectory divergence over many steps (Remark 7.2), but no theorem, corollary, or proof in this paper’s current form invokes it. We name it here rather than silently omit it.

C Full Proofs

Proof of Lemma 4.1. Induction on N . Base case ($N = 0$): $\Delta^0(L_i^{(t)} - L^*) = c_i r_i^t$. Inductive step. Assume $\Delta^{N-1}(L_i^{(t)} - L^*) = c_i r_i^{t-(N-1)}(r_i - 1)^{N-1}$. Then

$$\begin{aligned}\Delta^N(L_i^{(t)} - L^*) &= \Delta^{N-1}(L_i^{(t)} - L^*) - \Delta^{N-1}(L_i^{(t-1)} - L^*) \\ &= c_i r_i^{t-N+1}(r_i - 1)^{N-1} - c_i r_i^{t-N}(r_i - 1)^{N-1} \\ &= c_i(r_i - 1)^{N-1}(r_i^{t-N+1} - r_i^{t-N}) = c_i r_i^{t-N}(r_i - 1)^N,\end{aligned}$$

which is the claim for N . $\square$

Proof of Theorem 4.2. By Lemma 4.1, $\Delta^N(L_i^{(t)} - L^*) = c_i r_i^{t-N}(r_i - 1)^N$ and $\Delta^{N-1}(L_i^{(t)} - L^*) = c_i r_i^{t-N+1}(r_i - 1)^{N-1}$. Dividing,

$$\frac{\Delta^N(L_i^{(t)} - L^*)}{\Delta^{N-1}(L_i^{(t)} - L^*)} = \frac{r_i^{t-N}}{r_i^{t-N+1}} \cdot (r_i - 1) = r_i^{-1}(r_i - 1) = \frac{r_i - 1}{r_i},$$

where c_i cancels; the denominator is nonzero whenever $c_i > 0$, $r_i \in (0, 1)$, $N \geq 1$. $\square$

Proof of Proposition 4.3. Step 1 (pointwise perturbation). Let $f_1(r) = r^{t-1}(1-r)$, $f_2(r) = r^{t-2}(1-r)^2$ on $[\bar{r} - \delta_r, \bar{r} + \delta_r]$; both are smooth there, so $L_N := \sup_r |f'_N(r)| < \infty$. Writing $r_i = \bar{r} + \epsilon_i$, $|\epsilon_i| \leq \delta_r$, and $\Delta_i^{(N)} := f_N(r_i) - f_N(\bar{r})$, the mean value theorem gives $|\Delta_i^{(N)}| \leq L_N \delta_r$. With $Y_i^{(N)} := c_i f_N(\bar{r})$, $E_i^{(N)} := c_i \Delta_i^{(N)}$: $X_i(N) = Y_i^{(N)} + E_i^{(N)}$, $|E_i^{(N)}| \leq c_{\max} L_N \delta_r$.

Step 2 (mean perturbation). $|\mathbb{E}[X(N)] - \mathbb{E}[Y(N)]| \leq c_{\max} L_N \delta_r$.

Step 3 (variance perturbation). $X_i(N)^2 - Y_i(N)^2 = E_i^{(N)}(2Y_i^{(N)} + E_i^{(N)})$ is $O(\delta_r)$ uniformly, so $|\text{Var}(X(N)) - \text{Var}(Y(N))| \leq (A_N + B_N)\delta_r$ for finite A_N, B_N , hence $|\sqrt{\text{Var}(X(N))} - \sqrt{\text{Var}(Y(N))}| \leq D_N \delta_r$.

Step 4 (Lipschitz bound on CV). $\phi(\mu, s) = s/\mu$ is C^1 with bounded gradient Λ_N on the relevant compact region, so $|\text{CV}(X(N)) - \text{CV}(Y(N))| \leq \Lambda_N(c_{\max} L_N + D_N) \delta_r$.

Step 5. $\text{CV}(Y(N)) = \sqrt{\text{Var}(c)}/\mathbb{E}[c]$ is identical for $N = 1, 2$ ($f_N(\bar{r})$ cancels in the ratio); applying Step 4 to both and the triangle inequality gives the claim with $C := \max_N \Lambda_N(c_{\max} L_N + D_N)$, made explicit in Appendix F. $\square$

Proof of Theorem 5.1. By linearity, $\Delta^N \tilde{L}_i^{(t)} = a + b$ with $a := \Delta^N L_i^{(t)} = c_i r_i^{t-N}(r_i - 1)^N$ (deterministic, Lemma 4.1) and $b := \Delta^N \xi_{i,t} = \sum_{k=0}^N (-1)^k \binom{N}{k} \xi_{i,t-k}$. Then $\mathbb{E}[(\Delta^N \tilde{L}_i^{(t)})^2] = a^2 + 2a\mathbb{E}[b] + \mathbb{E}[b^2]$. By Assumption 3.3, $\mathbb{E}[b] = 0$; since the $\xi_{i,t-k}$ are independent with variance σ^2 , $\mathbb{E}[b^2] = \sigma^2 \sum_k \binom{N}{k}^2 = \sigma^2 \binom{2N}{N}$ (Vandermonde). Combining gives Equation (3). $\square$

Proof of Proposition 5.2. Write $\alpha_i := (1 - r_i)^2/r_i^2$, so $r_i^{2(t-N)}(r_i - 1)^{2N} = r_i^{2t} \alpha_i^N$ and $\text{MSE}(N, t) = c_i^2 r_i^{2t} \alpha_i^N + \binom{2N}{N} \sigma^2$. Using $\binom{2}{1} = 2$, $\binom{4}{2} = 6$, $\binom{6}{3} = 20$: $\text{MSE}(1, t) - \text{MSE}(2, t) = c_i^2 r_i^{2t} \alpha_i(1 - \alpha_i) - 4\sigma^2$, giving Equation (4); $\text{MSE}(3, t) - \text{MSE}(2, t) = 14\sigma^2 - c_i^2 r_i^{2t} \alpha_i^2(1 - \alpha_i)$, giving Equation (5). For $\alpha_i < 1$ ($r_i > \frac{1}{2}$), $(1 - \alpha_i) > 0$, so both impose genuine two-sided bounds combining into the band above (Equations (4) and (5)); non-emptiness holds since $4\alpha_i \leq 14$ for every $\alpha_i < 1$. $\square$

Proof of Corollary 5.3. By Assumption 3.3, $\text{Var}(\Delta^2 \xi_{i,t}) = \text{Var}(\xi_{i,t} - 2\xi_{i,t-1} + \xi_{i,t-2}) = \sigma^2(1 + 4 + 1) = 6\sigma^2$, matching Theorem 5.1 at $N = 2$. The Gaussian approximation is standard. Under Assumption 3.4, $\Delta^2 \xi_{i,t}$ is a linear combination of independent σ^2 -sub-Gaussian variables with squared coefficients summing to 6, hence $6\sigma^2$ -sub-Gaussian; the standard tail inequality gives $2 \exp(-k^2/2)$. $\square$

Proof of Theorem 6.1. Let $\Delta\theta^{(s)} = -\eta g^{(s)}$ (full-batch). Third-order Taylor expansion with remainder (Assumption B.2):

$$L_i(\theta^{(s)} + \Delta\theta^{(s)}) = L_i^{(s)} + \langle g_i^{(s)}, \Delta\theta^{(s)} \rangle + \frac{1}{2} \Delta\theta^{(s)\top} H_i(\theta^{(s)}) \Delta\theta^{(s)} + r_{i,s}, \quad |r_{i,s}| \leq \frac{M_3}{6} \eta^3 \|g^{(s)}\|^3.$$

Applying at $s = t-1, t-2$ and forming $\Delta^2 L_i^{(t)} = [L_i^{(t)} - L_i^{(t-1)}] - [L_i^{(t-1)} - L_i^{(t-2)}]$, the $O(\eta)$ terms combine into the drift term and the $O(\eta^2)$ terms into the curvature difference; $R_{i,t} := r_{i,t-1} - r_{i,t-2}$ and the triangle inequality give the stated remainder bound. $\square$

Proof of Theorem 7.1. At each $t \in [T-K, T]$ both runs use Equation (7) on the same reference trajectory, so the only difference between update directions is the term $\frac{1}{N} g_i^{(t)}$: $\|\theta_{\text{full}}^{(t+1)} - \theta_{\text{rem}}^{(t+1)}\| \leq \frac{\eta}{N} \|g_i^{(t)}\| \leq \frac{\eta\varepsilon}{N}$. Summing over K steps (triangle inequality) gives $\|\theta_{\text{full}}^{(T)} - \theta_{\text{rem}}^{(T)}\| \leq K\eta\varepsilon/N$; the test-loss bound follows from β_{test} -Lipschitzness. $\square$

Proof of Corollary 7.3. $\beta_{\text{test}} K\eta\varepsilon/N \leq \tau \iff K \leq N\tau/(\beta_{\text{test}}\eta\varepsilon)$; the largest integer satisfying this is $K_{\text{safe}} = \lfloor N\tau/(\beta_{\text{test}}\eta\varepsilon) \rfloor$ (a ceiling would round up past the constraint). $\square$

Proof of Lemma 7.7. By Λ -smoothness (descent lemma), $f(\theta - \frac{1}{\Lambda} \nabla f(\theta)) \leq f(\theta) - \frac{1}{2\Lambda} \|\nabla f(\theta)\|^2$. Combined with $f(\theta^*) \leq f(\theta - \frac{1}{\Lambda} \nabla f(\theta))$: $f(\theta^*) \leq f(\theta) - \frac{1}{2\Lambda} \|\nabla f(\theta)\|^2$, which rearranges to the claim. $\square$

Proof of Theorem 7.8. By Lemma 7.7, $\|g_i^{(t)}\|^2 \leq 2\Lambda_i(L_i^{(t)} - L^*) = 2\Lambda_i c_i r_i^t$ (Assumption 3.2). By Lemma 4.1 at $N = 2$, $\Delta^2 L_i^{(t)} = c_i r_i^{t-2} (1 - r_i)^2$, so $c_i r_i^t = (r_i/(1 - r_i))^2 \Delta^2 L_i^{(t)}$. Substituting gives $\|g_i^{(t)}\|^2 \leq 2\Lambda_i (r_i/(1 - r_i))^2 \Delta^2 L_i^{(t)}$, and taking square roots gives Equation (8). $\square$

Proof of Corollary 7.9. $\Delta^2 L_i^{(t)} = \hat{S}_i \mp \Delta^2 \xi_{i,t}$, so $\hat{S}_i < \delta$ with $|\Delta^2 \xi_{i,t}| \leq k\sqrt{6}\sigma$ implies $\Delta^2 L_i^{(t)} < \delta'$, hence $\|g_i^{(t)}\| < \varepsilon_i(\delta')$ by Theorem 7.8. So the event in the statement is contained in $\{|\Delta^2 \xi_{i,t}| > k\sqrt{6}\sigma\}$, bounded by Corollary 5.3 at $2\exp(-k^2/2)$. $\square$

D Toward a Multi-Step Trajectory-Divergence Bound

Remark 7.2 names this paper’s central open theoretical question: Theorem 7.1 bounds only a single removal decision’s instantaneous perturbation on a shared reference trajectory, not how far Algorithm 1’s actual, multi-removal trajectory drifts from the full-data one over an entire run. This appendix reports a first, honest attempt using only assumptions already in this paper.

Setup. Let θ_t^{full} follow full-batch GD on the population loss, $\theta_{t+1}^{\text{full}} = \theta_t^{\text{full}} - \eta g^{\text{full}}(\theta_t^{\text{full}})$, $g^{\text{full}}(\theta) := \frac{1}{N} \sum_i g_i(\theta)$, and $\theta_t^{\text{TG-IES}}$ follow Algorithm 1’s actual update, $\theta_{t+1}^{\text{TG-IES}} = \theta_t^{\text{TG-IES}} - \eta \hat{g}_t(\theta_t^{\text{TG-IES}})$, $\hat{g}_t(\theta) := \frac{1}{N} \sum_{i \in A_t} g_i(\theta)$, where A_t is the algorithm’s own, evolving active set. Both start at θ_0 ; define $e_t := \theta_t^{\text{full}} - \theta_t^{\text{TG-IES}}$, $e_0 = 0$.

Proposition D.1 (Exponential trajectory-divergence bound). Suppose Assumption 3.1 holds, and every excluded instance $i \notin A_t$ satisfies $\|g_i(\theta_t^{\text{TG-IES}})\| < \varepsilon_i$ at every step t along the algorithm’s actual trajectory. Let $\bar{\varepsilon}_t := \frac{1}{N} \sum_{i \notin A_t} \varepsilon_i$, $\varepsilon_{\max} := \max_i \varepsilon_i$. Then

$$\|e_t\| \leq \eta \sum_{s=0}^{t-1} (1 + \eta\beta)^{t-1-s} \bar{\varepsilon}_s \leq \frac{\varepsilon_{\max}}{\beta} \left[(1 + \eta\beta)^t - 1 \right]. \quad (9)$$

Splitting the per-step gradient difference into a same-map term (bounded by β -smoothness) and a set-difference term (bounded by the gradient-floor hypothesis), a triangle-inequality recursion in $\|e_t\|$ unrolls to this closed form. The bound is exponential in t and vacuous over a full ~ 200 -epoch/many-thousand-step run for any realistic β : useful precisely because it is a correct negative-ish result, explaining mechanistically why a smoothness-only (Lipschitz, non-contractive) argument cannot give a practically useful multi-step bound, rather than merely asserting the gap is hard.

Remark D.2 (What would fix this). Section 10.2’s measurements show absolute divergence plateauing rather than growing without bound – qualitatively inconsistent with Proposition D.1’s exponential worst case, consistent instead with a contractive map. If L were additionally μ -strongly convex (or satisfied a local Polyak-Łojasiewicz condition) along the trajectory, the GD map is a contraction with factor $\rho = \max(|1 - \eta\mu|, |1 - \eta\beta|) < 1$, replacing this appendix’s $(1 + \eta\beta)$ expansion factor with $\rho < 1$ in the same recursion, giving a bounded, t -independent limit $\eta\varepsilon_{\max}/(1 - \rho)$ instead of exponential growth – qualitatively matching Section 10.2’s plateau. This needs (i) adapting the standard co-coercivity argument to the g^{full} -vs- $\hat{g}_t$ split, not just the textbook two-point case, and (ii) justifying local strong convexity or a PL condition along a deep network’s actual trajectory – an assumption with real literature behind it in restricted regimes (late-training, over-parameterized, near a broad minimum) but one we have not verified for this paper’s architectures. We flag this as the concrete next step rather than derive it here.

E An Aggregate Refinement: Population-Level Gradient Cancellation

Proposition D.1 bounds the per-step set-difference term via the triangle inequality over per-instance gradient-norm bounds ε_i – exactly the hypothesis Appendix T finds violated by roughly half the excluded set at every snapshot. This section derives what that term actually is, shows the triangle-inequality step is provably loose whenever excluded gradients do not all point the same way, and measures the resulting slack directly, addressing Appendix T’s “natural but unchecked candidate” reconciliation and Appendix U’s item (5).

An exact identity. Write $R_t := D \setminus A_t$ for the excluded set at step t , $\pi_t := |R_t|/N \in [0, 1]$ for the exclusion fraction (already logged every epoch as `total_excluded_now/N`), and $\mu_t(\theta) := \frac{1}{|R_t|} \sum_{i \in R_t} g_i(\theta)$ for the mean gradient of the excluded set ($\mathbf{0}$ if $R_t = \emptyset$). Then, exactly, no approximation:

$$g^{\text{full}}(\theta) - \hat{g}_t(\theta) = \frac{1}{N} \sum_{i \in R_t} g_i(\theta) = \pi_t \mu_t(\theta). \quad (10)$$

Lemma E.1 (The per-instance bound is a relaxation of the true term). For any θ , $\|\mu_t(\theta)\| \leq \frac{1}{|R_t|} \sum_{i \in R_t} \|g_i(\theta)\|$ (triangle inequality / Jensen), with equality iff every $g_i(\theta)$, $i \in R_t$, points in exactly the same direction. If $\|g_i(\theta)\| < \varepsilon_i$ for all $i \in R_t$, then $\pi_t \|\mu_t(\theta)\| \leq \pi_t \cdot \frac{1}{|R_t|} \sum_{i \in R_t} \varepsilon_i = \bar{\varepsilon}_t$, Proposition D.1’s own quantity.

This is not a new assumption on top of Proposition D.1 – it shows that bound already over-approximates the true recursion input $\pi_t \mu_t(\theta)$ by exactly the amount excluded gradients cancel directionally, and that this cancellation is measurable directly, without ever estimating individual ε_i .

Assumption E.2 (Aggregate gradient floor). There exists $\bar{\varepsilon} \geq 0$ with $\|\mu_t(\theta_t^{\text{TG-IES}})\| \leq \bar{\varepsilon}$ for every step t along the algorithm’s actual trajectory.

Unlike Proposition D.1’s per-instance hypothesis, Assumption E.2 does not require any individual excluded gradient to be small – only that the excluded set’s gradients do not all point the same way strongly enough to add up. It can hold when the per-instance hypothesis fails on half the excluded set (Appendix T); it could in principle fail even when every individual gradient is small, if they happened to be exactly aligned, though nothing in the measurement below suggests that is what happens here.

Proposition E.3 (Aggregate-floor multi-step bound). Suppose Assumptions 3.1 and E.2 hold, and let $\bar{\pi} := \max_t \pi_t$. Then, by the same same-map/set-difference split and triangle-inequality recursion as Proposition D.1 (only the set-difference term’s bound changes, via Equation (10)),

$$\|e_t\| \leq \eta \bar{\pi} \bar{\varepsilon} \sum_{s=0}^{t-1} (1 + \eta\beta)^{t-1-s} = \frac{\bar{\pi} \bar{\varepsilon}}{\beta} \left[(1 + \eta\beta)^t - 1 \right]. \quad (11)$$

This is the same exponential form as Proposition D.1: Lemma E.1 shows the leading constant can only shrink ($\bar{\pi} \bar{\varepsilon} \leq \max_t \bar{\varepsilon}_t$), never grow, but the expansion factor $(1 + \eta\beta)^t$ is unchanged,

so this does not resolve Remark D.2’s structural vacuousness over hundreds of epochs by itself – its value is a tighter, directly-measurable leading constant, not a qualitatively different bound.

Corollary E.4 (Aggregate conditional exclusion bound). Under the shared-reference idealization of Theorem 7.1 applied to the whole excluded set R_t rather than one instance, tolerance τ is kept ($|L_{\text{test}}(\theta_{\text{full}}^{(T)}) - L_{\text{test}}(\theta_{\text{rem}}^{(T)})| \leq \tau$) for any $K \leq K_{\text{safe}}^{\text{agg}} := \lfloor \tau / (\beta_{\text{test}} \eta \bar{\pi} \bar{\varepsilon}) \rfloor$ – Corollary 7.3’s formula without the factor of N , since $\bar{\varepsilon}$ is already a per-capita quantity rather than one raw gradient norm.

Direct measurement. We extend the reactivation probe (Appendix T) to also accumulate, alongside each probed instance’s individual gradient, the running sum of the raw gradient vectors across the 300-instance probe sample, reporting $\|\mu_t\|$ (computed on the probe subsample, an estimate of the true population quantity) next to the already-reported $\text{mean}_i \|g_i^{(t)}\|$ (primary cell, self-calibrated δ , 3 seeds, tagged `_reactprobe2`; identical training run to Appendix T’s `_reactprobe`, purely additional instrumentation – accuracy/savings match to within run-to-run noise, confirming the extra accumulation does not perturb training).

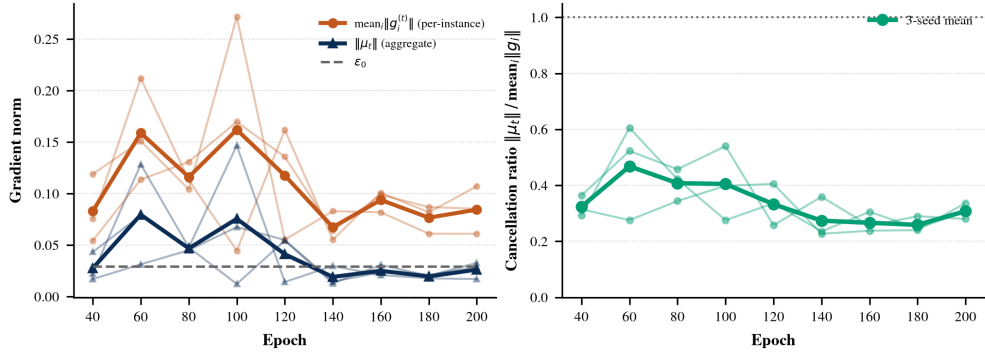


Figure 2: Left: $\text{mean}_i \|g_i^{(t)}\|$ (per-instance) vs. $\|\mu_t\|$ (aggregate) vs. the calibrated floor ε_0 , primary cell, 3 seeds (light: per-seed; bold: 3-seed mean). Right: the cancellation ratio $\|\mu_t\| / \text{mean}_i \|g_i^{(t)}\|$ at each probed epoch.

Finding: partial cancellation is real, but the resulting bound still does not cover this paper’s own K^* . Pooling all 9 probed epochs across 3 seeds (27 (epoch, seed) points; accuracy/savings for these runs, $94.73 \pm 0.02\%$ / $53.66 \pm 0.09\%$, match Table 3’s unprobed TG-IES numbers closely, confirming this extra instrumentation does not perturb training), the mean cancellation ratio is $\|\mu_t\| / \text{mean}_i \|g_i^{(t)}\| = 0.34$, ranging 0.26–0.47 epoch-to-epoch (Figure 2) – real, substantial directional cancellation (Lemma E.1’s inequality is far from tight, not saturated), consistent with the “diluted by averaging” half of Appendix T’s previously-unverified reconciliation candidate. But cancellation alone is not the same as a small aggregate, and Corollary E.4’s own honest, sup-based constants tell a less comforting story: using the single largest observed $\|\mu_t\|$ across the whole horizon ($\bar{\varepsilon} = 0.147$) and the largest observed exclusion fraction ($\bar{\pi} = 0.881$) – the same conservative floor-over-a-horizon convention Table 1 already uses for the per-instance bound – gives $K_{\text{safe}}^{\text{agg}} = \lfloor 5.5 \rfloor = 5$, smaller than the $K^* = 15$ ceiling used in every run in this paper. Unlike the per-instance K_{safe} (Remark 7.5, $\approx 1.2\text{M}$, wildly non-binding), the aggregate refinement is tight enough to be genuinely violated at this paper’s own configuration, not merely slack. This is not a contradiction with Section 10.2’s empirical plateau: Proposition E.3’s bound, like Proposition D.1’s, is still exponential in t and built by the same worst-case triangle-inequality recursion, so a bound constructed to hold under adversarial per-step alignment can be loose enough to be violated on paper while the one, actually-realized trajectory still stays bounded – exactly Appendix D’s “correct but vacuous” story, now reproduced at the aggregate level with a measured, not merely asserted, constant. We read this as narrowing Remark 7.4’s gap, not closing it: the cancellation this section set out to check for is real and directionally as hypothesized, but it is not large enough, combined with this paper’s actual exclusion fractions, for Corollary E.4’s

worst-case bound to cover the patience this paper actually uses – only the direct trajectory instrumentation of Section 10.2 does that, empirically rather than via a bound.

F Explicit Constants for Proposition 4.3

Recall from Assumption B.1 that $\text{Var}(c) \leq (c_{\max} - c_{\min})^2/4 =: V_c$. One concrete choice of the constant in Proposition 4.3 is

$$C(t, \bar{r}, c_{\min}, c_{\max}) := \max_{N \in \{1, 2\}} \left(\frac{c_{\max} L_N}{c_{\min} f_N(\bar{r})} + \frac{c_{\max} L_N}{\sqrt{V_c} c_{\min} f_N(\bar{r}) + 10^{-12}} \right),$$

where $f_1(r) = r^{t-1}(1-r)$, $f_2(r) = r^{t-2}(1-r)^2$, $L_N = \sup_{r \in [\bar{r}-\delta_r, \bar{r}+\delta_r]} |f'_N(r)|$. The 10^{-12} guards against division by zero when all c_i are equal ($V_c = 0$).

G Counterexample: Global CV Monotonicity Can Fail

A global claim $\text{CV}(\Delta^2 L) \leq \text{CV}(\Delta^1 L)$ (without Assumption 3.5) can fail under wide rate dispersion. Take $c_1 = c_2 = 1$, $L^* = 0$, $r_1 = 0.01$, $r_2 = 0.99$, $t = 10$:

$$\Delta^1(L_1^{(10)}) \approx -10^{-18}, \quad \Delta^1(L_2^{(10)}) \approx -0.0091, \quad \Delta^2(L_1^{(10)}) \approx 10^{-16}, \quad \Delta^2(L_2^{(10)}) \approx 9.2 \times 10^{-5}.$$

The cross-instance magnitude ratio is order 10^{16} at $N=1$ and order 10^{11} at $N=2$: dispersion (CV across the two instances) increases, not decreases, from $N=1$ to $N=2$ here, confirming Assumption 3.5 is an important assumption, not a technicality.

H Extended Practical Implementation Notes

Recalibration schedule. δ and K^* are recalibrated once, at epoch 30, then held fixed, rather than repeating every Δ epochs: repeating was found counterproductive, since a shrinking noise floor $\hat{\sigma}$ re-tightens δ over time and every recalibration resets in-progress K^* -window counts, together strangling savings. This is an empirical finding, not a theorem (Section 12).

Gradient-norm floor estimate. ε_0 is a low percentile (default 20th) of a cheap last-layer proxy $\|\text{softmax}(z_i) - \text{onehot}(y_i)\| \cdot \|\text{feat}_i\|$ over a subsample of the active set (capped at 8192 instances), cutting recalibration cost roughly $6\times$ versus a full backward pass without materially changing the percentile estimate.

Consistent forward-pass discipline. Following Yuan et al. (2025)’s stated Algorithm 1 rather than the shipped implementation, all instances (active and passive) are forwarded in the same `train()` mode every epoch (passive under `no_grad`), avoiding a train/eval batch-normalization mismatch (Ioffe & Szegedy, 2015).

A rounding-direction correction, with no effect on reported results. Corollary 7.3 rounds K_{safe} down; the implementation generating every run in this paper used a ceiling instead, caught only while finalizing this version. Checking every run’s logged K^* trajectory: in every case K^* reaches `max_kstar` = 15 immediately upon recalibration, i.e., the uncapped value already exceeds the clamp before rounding is applied, so ceiling vs. floor changes nothing here. The implementation now uses the floor consistent with Corollary 7.3.

Normalization: $1/N$ vs. $1/|A_t|$ is not a live ablation under this paper’s configuration. Equation (7)’s $1/N$ -scaled active-set gradient and shipped IES’s $1/|A_t|$ mean differ, in principle, whenever a training step processes fewer than N active instances at once. Every method here trains with fixed-size minibatches (`batch_size` = 64) and `drop_last` = True whenever the active set is at least one minibatch, so every optimizer step in every run sees a full batch of exactly 64 instances, and the per-step loss (`losses.sum()/batch_size` for TG-IES, `losses.mean()` for the other three methods) is then numerically identical between the two normalizations; `drop_last` discards a final undersized batch outright rather than processing it under either convention. We checked this holds throughout, not just argued it: across all 44 distinct TG-IES runs in this paper, the smallest active-set size ever reached is 2,965 (VGG-16, seed 1), $46\times$ `batch_size`, so no run here ever executes a step where the two normalizations could numerically differ. Running a normalization ablation under this paper’s own configuration would therefore measure exactly zero effect by construction, not because

the choice is unimportant to the theory – Equation (7) is exactly what Theorem 7.1’s bound needs (Section 7) – but because no run here ever reaches the regime (active set below one minibatch) where the two formulas’ predictions part ways. Testing that would need a setting engineered to push the active set below batch_size (a far more aggressive K^* -sweep than Section 10.2’s, or a much smaller batch_size), which we have not run.

I Full Zero-Shot Transfer Table

Table 6: TG-IES, every distinct (dataset, architecture, optimizer/schedule) cell in this paper, same four fixed calibration constants throughout. Mean $\pm$ std over seeds where $n > 1$; asterisked rows are single-seed by design (Appendix K’s matched-epoch schedule comparison).

Cell	n	Acc. (%)	Saved (%)
CIFAR-10 / ResNet-18 / SGD-exp (primary)	3	94.85 $\pm$ 0.28	53.63 $\pm$ 0.30
CIFAR-10 / ResNet-18 / SGD-fixed	3	91.91 $\pm$ 0.18	29.27 $\pm$ 0.30
CIFAR-10 / ResNet-18 / SGD-linear	1	94.85*	32.05*
CIFAR-10 / ResNet-18 / SGD-step	1	94.44*	43.52*
CIFAR-10 / ResNet-18 / SGD-cosine	1	94.73*	23.15*
CIFAR-10 / ResNet-18 / Adam	3	92.44 $\pm$ 0.10	30.18 $\pm$ 0.62
CIFAR-10 / VGG-16 / SGD-exp	3	93.11 $\pm$ 0.14	54.43 $\pm$ 0.12
CIFAR-10 / DenseNet-121 / SGD-exp	3	95.28 $\pm$ 0.09	54.53 $\pm$ 0.66
CIFAR-100 / ResNet-18 / SGD-exp	3	76.01 $\pm$ 0.08	57.76 $\pm$ 0.04
CIFAR-10 / ViT-Tiny / AdamW-wc	3	83.71 $\pm$ 0.47	25.52 $\pm$ 0.26
CIFAR-100 / ViT-Tiny / AdamW-wc	3	58.56 $\pm$ 0.47	26.28 $\pm$ 0.14

The savings figure varies substantially across cells (23–58%), as it should: schedule-, dataset-, and architecture-dependent mechanisms (Appendices J, K and Q) directly cause this variation. What does not vary is the calibration procedure: this table was assembled with no per-cell constant lookup, because there is none to look up.

J Second Optimizer: Adam

Table 7: ResNet-18, CIFAR-10, Adam, mean $\pm$ std, 3 seeds.

Method	Test Acc. (%)	Backprop Saved (%)
Baseline	92.91 $\pm$ 0.06	0.03 $\pm$ 0.00
IES (shipped)	92.61 $\pm$ 0.10	56.02 $\pm$ 0.22
IES (literal)	92.42 $\pm$ 0.10	60.09 $\pm$ 0.29
TG-IES (ours)	92.44 $\pm$ 0.10	30.18 $\pm$ 0.62

Adam (Kingma & Ba, 2015) is a second, more pronounced instance of the fixed-LR-style pattern of Section 9: both IES variants save more (56.0–60.1%) than TG-IES’s 30.2%, at accuracy within 0.02–0.17 points of TG-IES’s (Appendix W: the accuracy gap vs. IES (literal) is not distinguishable from zero at $n = 3$, while the saved-% deficit is consistent in sign across all 3 seeds). Adam and fixed-LR SGD share the structural property distinguishing them from every cell where TG-IES leads: neither has the exponential schedule’s large, steadily-decaying learning rate. We report this as an open empirical pattern, not an explained one; Appendix K proposes and checks a candidate mechanism against the logged record.

K Toward a Mechanism: The Drift Term Under Non-Decaying Learning Rates

Theorem 6.1 decomposes $\Delta^2 L_i^{(t)} = \eta_t D_i^{(t)} + O(\eta_t^2) + R_{i,t}$, $D_i^{(t)} := \langle g_i^{(t-2)}, g^{(t-2)} \rangle - \langle g_i^{(t-1)}, g^{(t-1)} \rangle$. Under the exponential schedule, $\eta_t \rightarrow 0$ drives the whole drift term toward zero regardless of $D_i^{(t)}$ ’s own decay; under a non-decaying schedule (fixed-LR SGD, or Adam’s lack of scalar- η decay), only $D_i^{(t)} \rightarrow 0$ on its own can shrink it, which Assumption 3.2’s geometric picture gives no particular reason to expect on any fixed timetable. This predicts δ should calibrate smaller wherever η is smaller, and fewer instances should reach $K^*=15$ consecutive low-signal epochs. Both hold in the logged data: at epoch 31, δ recalibrates to 0.01597 (exponential, $\eta_{31} \approx 0.0282$) vs. 0.00247 (fixed-LR, $6.5\times$ smaller) and

0.000257 (Adam, $62\times$ smaller); by training’s end the exponential run excludes 87.4% of instances vs. 42.0% (fixed-LR) and 45.6% (Adam), matching the backprop-saved gap.

Direct instrumentation of $D_i^{(t)}$. A diagnostic run computing $g^{(t)}$, every $L_i^{(t)}$, and $\langle g_i^{(t)}, g^{(t)} \rangle$ for 300 fixed instances, baseline training, all three schedules, complicates the “drift dominates” reading: $\eta_t D_i^{(t)}$ correlates only weakly with the observed $\Delta^2 L_i^{(t)}$ (Pearson $r = 0.34, 0.37, 0.23$ for exponential/fixed/Adam), and its magnitude does not dominate the combined curvature-plus-remainder term (ratio $0.69\times, 0.32\times$, and effectively $0\times$ respectively) – the opposite of a literal “drift dominates” reading, and it weakens further in the last training quarter (Figure 3). We read this as refining, not refuting, Theorem 6.1: the $O(\eta)$ -vs- $O(\eta^2)$ ordering is asymptotic as $\eta \rightarrow 0$, and at this network’s actual step sizes (η up to 0.1) the curvature term is evidently not negligible in absolute terms. The qualitative argument above only requires drift to scale with η_t , which this does not contradict; but $\Delta^2 L_i^{(t)}$ being well-approximated by $\eta_t D_i^{(t)}$ alone is not supported, and Section 6’s framing is corrected accordingly.

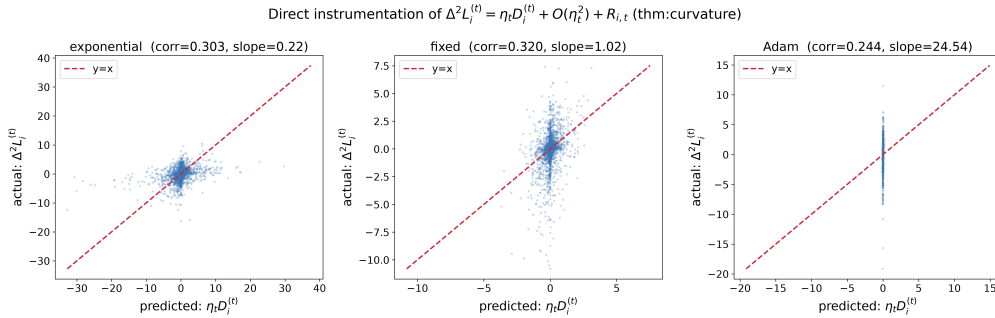


Figure 3: Direct instrumentation of $\Delta^2 L_i^{(t)} = \eta_t D_i^{(t)} + O(\eta_t^2) + R_{i,t}$, fixed 300-instance subset, baseline training, all valid epochs, one panel per schedule. Dashed line: $y = x$. The drift term correlates weakly with the observed signal and does not dominate the residual under any schedule.

L Full New-Baselines Comparison and Fixed-Compute-Budget Analysis

We implement and run four baselines not previously benchmarked against IES: InfoBatch (Qin et al., 2024) ($r=0.5$ soft-pruning below epoch-mean loss, $1/(1-r)$ gradient rescaling, annealing to the full dataset for the final 12.5% of epochs); BLS-InfoBatch (Zhou et al., 2026) (same schedule, EMA-smoothed score, $\alpha=0.7$); AlignPrune (Qin et al., 2026) (Dynamic Alignment Score against the population-mean trajectory, since no clean-labeled reference subset exists here); and Selective Backprop (Jiang et al., 2019) (percentile-rank accept probability $p^{\beta_{sb}-1}$, $\beta_{sb}=3$, on the persistent last-observed-loss score, no gradient rescaling and no annealing back to the full dataset, since SB is an explicitly biased sampler by design rather than InfoBatch’s unbiased correction). We do not compare against DPPO (Zhu et al., 2026) (a different, RL setting).

On VGG-16 (seed 0) and CIFAR-100 (seed 0), the same ranking as the primary cell holds among the first three: TG-IES leads at 54.4%/57.8% saved against InfoBatch’s 38.9%/35.2%, BLS-InfoBatch’s 36.4%/30.9%, and AlignPrune’s 22.7%/22.5%, all within 1–2 accuracy points. AlignPrune saves the least of these methods consistently, expected rather than a weakness of our implementation: its Dynamic Alignment Score identifies samples misaligned with the population trend (a label-noise signal), not samples converged with it (a mastery signal); on clean CIFAR labels most of what it would need to prune to match TG-IES are exactly the well-aligned instances its score is built to keep.

Selective Backprop: the other extreme of the trade-off. On the primary cell (3 seeds), SB saves $66.72 \pm 0.01\%$ of backpropagation instances – more than every other method in this paper, TG-IES included (53.63%) – but at $93.31 \pm 0.11\%$ accuracy, a 1.72-point cost relative to baseline (95.03%), the largest accuracy cost of any method compared anywhere in this paper (next largest: TG-IES’s own 0.87 points on CIFAR-100, Section 10). SB’s savings are also the most seed-stable of any method measured (± 0.01 points, tighter even than InfoBatch’s ± 0.01). This is not a failure of our implementation: SB’s probabilistic accept/reject rule

has no calibrated stopping condition tied to a stability argument (Section 2), so it has no mechanism analogous to Corollary 7.3’s conditional exclusion bound to stop pruning before accuracy degrades. Read together with Table 3, SB and TG-IES mark the two ends of a Pareto-like frontier on this cell: the most raw compute saved for the most accuracy given up, versus a small, bounded accuracy cost for a smaller but still leading (among methods with any comparable accuracy) compute saving.

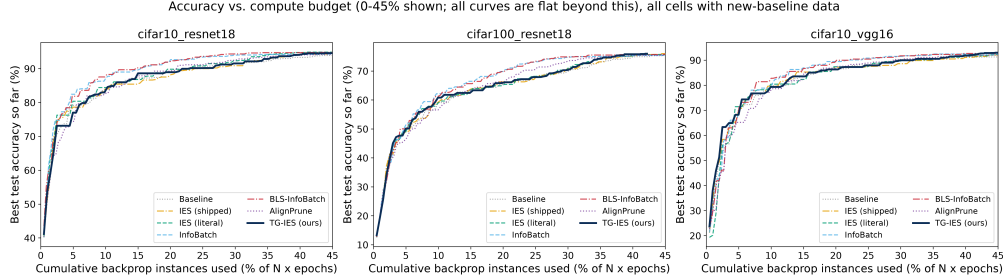


Figure 4: Best-so-far test accuracy vs. cumulative compute budget (% of $N \times \text{total-epochs}$), all three cells with new-baseline data. A method’s curve stops at its own final compute point.

Accuracy under a fixed compute budget. At a common compute budget rather than each method’s own stopping point, the result is a genuine trade-off, consistent across all three cells: at low budgets (20–30%), InfoBatch and BLS-InfoBatch lead TG-IES by 2.5–4.3 points, since they skip both forward and backward on a pruned instance and so get more model updates per unit of measured compute early in training. This narrows to 0.1–0.8 points by 40% and reverses on CIFAR-100 (+0.12 points for TG-IES). The practical implication is budget-dependent: a practitioner constrained to 20–30% of full compute would currently do better with InfoBatch-family methods; Table 3’s own-budget comparison describes a different, later operating regime. We report both framings rather than hide the trade-off.

M Empirical Validation of the Noise-Independence Assumption

Assumption 3.3 assumes $\xi_{i,t}$ independent across t ; we test this directly. Detrending each instance’s epoch-101–200 loss trajectory (ResNet-18/CIFAR-10, seed 0) with an OLS linear fit and pooling residuals across instances estimates lag- k autocorrelation directly (Figure 5).

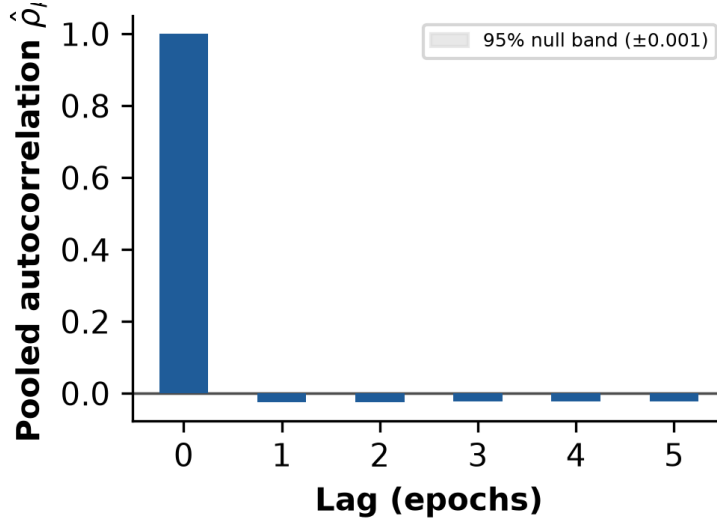


Figure 5: Pooled lag- k autocorrelation of detrended per-instance loss residuals, epochs 101–200, ResNet-18/CIFAR-10 seed 0 ($n \approx 1.9\text{M}$ pairs/lag). Shaded band: 95% null interval for zero correlation.

An earlier pass detrended with a width-5 moving average and measured $\hat{\rho}_1 = -0.298, \hat{\rho}_2 = -0.348$, which looked like strong evidence against Assumption 3.3; we verified by direct calculation that pure white noise under that same filter produces

$\rho_1 = -0.30, \rho_2 = -0.35$ exactly – the entire signal was the detrending filter’s own artifact. Redone with OLS linear detrending (whose own simulated white-noise artifact is $\hat{\rho}_1 \approx -0.021, \hat{\rho}_2 \approx -0.019$), the real data gives $\hat{\rho}_1 = -0.025, \hat{\rho}_2 = -0.023$ – statistically distinguishable from zero at $n \approx 1.9\text{M}$ pairs but small and close to the detrending method’s own bias, evidence for, not against, Assumption 3.3 once the method’s bias is accounted for. Taking the measured values at face value regardless, the correction to Theorem 5.1 at $N=2$ is a factor of 6.15 against the assumed 6.0 – a 2.5% correction, requiring δ to grow under 1.3% to keep the same nominal false-exclusion rate. This does not materially change any result in Section 9.

N Gradient-Norm-Floor Percentile Sensitivity

Section 7.3 fixes the percentile behind K^* at the 20th throughout; we sweep $\{5, 10, 20, 35, 50, 80\}$ on ResNet-18 and VGG-16 (primary recipe, seed 0), holding everything else fixed.

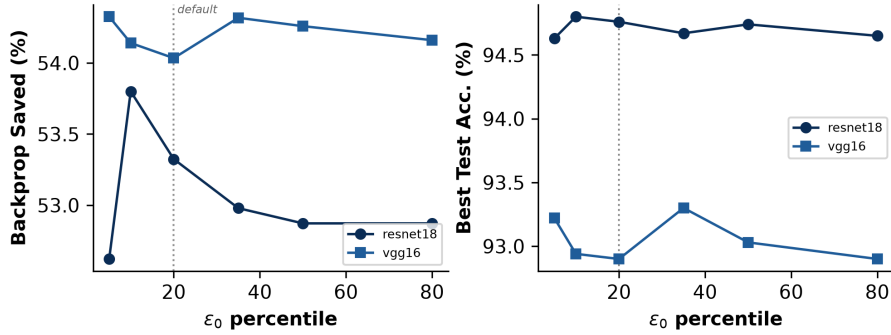


Figure 6: Backprop saved (left) and best test accuracy (right) vs. gradient-norm-floor percentile, ResNet-18 and VGG-16, CIFAR-10, seed 0. Dotted line: this paper’s fixed default (20th percentile).

Both curves are essentially flat across a $16\times$ range: ResNet-18 savings vary by 1.18 points, accuracy by 0.17; VGG-16 by 0.30 and 0.40 points respectively. Neither curve is monotonic, itself informative: there is no directional trend to correct for by tuning further. The gap between architectures at any fixed percentile is larger than the gap the percentile choice induces within either, consistent with the percentile being a fixed, low-sensitivity policy input rather than one needing per-architecture retuning within this range. This does not resolve Remark 7.4 (evidence the current value is not fragile, not evidence a derived value would not be preferable in principle).

O Measured Wall-Clock Time, Peak Memory, and Estimated Energy

Table 8: Measured efficiency, primary setting, single RTX 5090. Wall-clock: mean over 3 seeds from the real 200-epoch runs. Peak Mem/Util/Energy: short instrumented runs, seed 0; Energy is an estimate. Lower is better throughout.

Method	Wall-clk (min)	Peak Mem (MiB)	Util (%)	Energy (Wh)
Baseline	15.66	1162	67.7	98.2
IES (shipped)	12.86	1216	62.8	78.1
IES (literal)	12.26	1184	59.0	70.1
TG-IES	11.66	1454	48.9	59.9

Wall-clock is the per-epoch time already logged by every 200-epoch run; peak memory, mean utilization, and mean power were collected separately via out-of-process nvidia-smi polling on fresh 8-epoch instrumented repetitions; estimated energy multiplies each method’s measured mean power by its real 200-epoch wall-clock. TG-IES is fastest and most estimated-energy-efficient (25.5% faster, $\sim 39\%$ lower energy than baseline) but has the highest peak memory (1454 MiB, $\sim 25\%$ above baseline), attributable specifically to its one-time recalibration pass: estimating the gradient-norm-floor percentile forwards a subsample in batches of 256 (four times the training batch size), and PyTorch’s caching allocator does not release the resulting larger reservation for the rest of the run. This is a genuine memory/compute trade-off specific to TG-IES’s calibration mechanism.

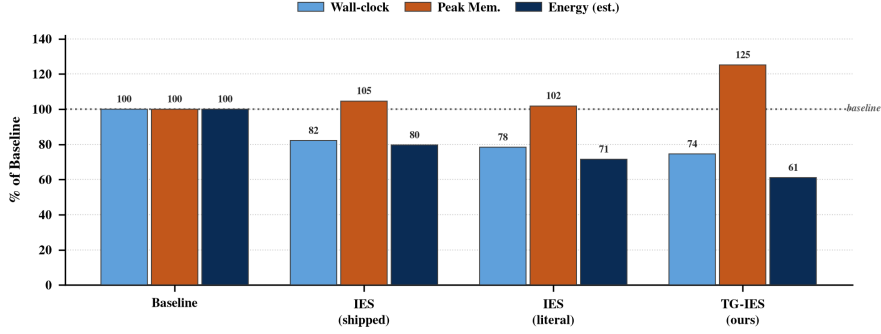


Figure 7: Wall-clock time, peak GPU memory, and estimated energy, each as % of no-removal baseline. Wall-clock and energy fall furthest for TG-IES; peak memory is the one metric that rises above baseline, only for TG-IES.

Does this generalize across batch size? Table 8 reports one batch size (64) on one GPU. We reuse a short instrumented sweep over {32, 64, 128, 256} (baseline, IES shipped, TG-IES; 8 epochs each, same nvidia-smi-polling methodology) to check.

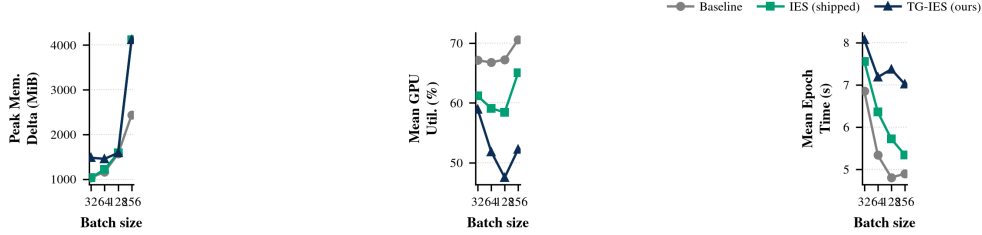


Figure 8: Peak GPU memory delta, mean GPU utilization, and mean per-epoch time (short 8-epoch instrumented runs, log-scale batch size) vs. batch size, ResNet-18/CIFAR-10.

This does not generalize the way Table 8’s headline number suggests. Within these short 8-epoch windows, before enough instances have been excluded to amortize TG-IES’s per-epoch bookkeeping, TG-IES is slower than baseline at every batch size tested (18–53% higher mean epoch time), the opposite direction of the full-200-epoch table’s 25.5% speedup. Table 8’s wall-clock advantage is consequently a full-run property that emerges only as exclusions accumulate over training, not a per-epoch constant – it should not be assumed to hold for short training runs or fine-tuning regimes where few instances ever reach K^* consecutive low-signal epochs. Peak-memory overhead relative to baseline is also batch-size-dependent and non-monotonic (1.01–1.70 $\times$ across the four sizes, both IES methods jumping to 4122MiB together at batch size 256, plausibly a CuDNN algorithm-selection effect at that size rather than anything specific to TG-IES’s own recalibration pass, though we have not isolated the cause). GPU utilization is lower for TG-IES than baseline at every batch size and the gap widens with batch size (8-point gap at 32, 23-point gap at 256) – consistent with the per-instance active/passive bookkeeping being harder to amortize as more images share one kernel launch. We report this as a genuine caveat on Table 8’s generalization, not something Table 8 itself is wrong about (that table’s numbers are for the real, full 200-epoch runs this paper’s accuracy/savings numbers also come from).

P Additional Figures for Level-2 and the Gradient-Norm Bridge

Q ViT-Tiny: Full Results and Caveats

Every architecture in the main matrix is convolutional; here we test whether TG-IES’s calibration – derived under no architectural assumption beyond a per-instance loss and gradient – transfers to a Vision Transformer (Dosovitskiy et al., 2021), trained from scratch at CIFAR scale with the same four fixed calibration constants as every convolutional cell. We use a CIFAR-native ViT-Tiny: patch-4 on the 32 $\times$ 32 input (64 tokens plus CLS), 6 pre-norm blocks, 192-dim, 3 heads, 2.69M parameters – not the ImageNet-oriented patch-

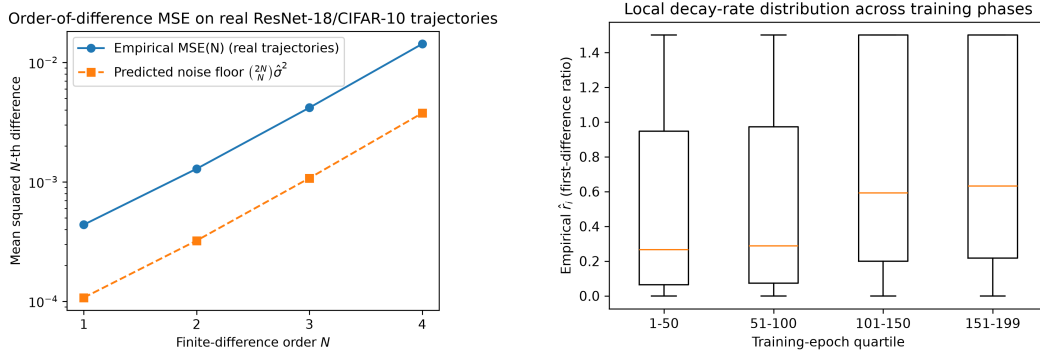


Figure 9: Left: empirical $MSE(N)$ vs. predicted noise floor, naive population-average test (all instances, epochs 101–200). This pooled test does not isolate Proposition 5.2’s SNR band and finds $N=1$ lowest; the band-restricted test (main text) finds $N=2$ lowest, as predicted. Right: empirical $\hat{r}_i$ pooled per training-epoch quartile; rate rises and spread narrows in the second half of training.

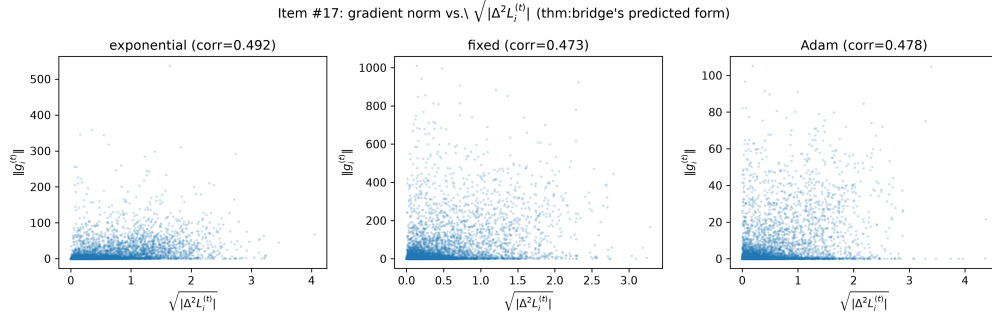


Figure 10: Per-instance gradient norm $\|g_i^{(t)}\|$ vs. $\sqrt{|\Delta^2 L_i^{(t)}|}$, all three LR schedules, primary architecture/dataset (300-instance subset). Theorem 7.8 predicts a positive, roughly linear relationship; correlations shown per panel.

16 configuration, whose tokenization would yield only 4 tokens at this resolution. AdamW (lr 10^{-3} , weight decay 0.05), 10-epoch linear warmup then cosine decay over 200 epochs; we validated the baseline alone before running the full comparison.

Table 9: ViT-Tiny (CIFAR-native, patch-4), CIFAR-10 and CIFAR-100, AdamW warmup-cosine, mean $\pm$ std, 3 seeds.

Method	CIFAR-10		CIFAR-100	
	Acc. (%)	Saved (%)	Acc. (%)	Saved (%)
Baseline	84.07 ± 0.35	0.03 ± 0.00	59.25 ± 0.24	0.03 ± 0.00
IES (shipped)	83.92 ± 0.18	13.13 ± 0.21	59.19 ± 0.22	8.38 ± 0.22
IES (literal)	83.86 ± 0.48	16.68 ± 0.32	58.77 ± 0.24	12.51 ± 0.21
TG-IES (ours)	83.71 ± 0.47	25.52 ± 0.26	58.56 ± 0.47	26.28 ± 0.14

Four findings. The savings advantage transfers and grows with difficulty: $1.53\times$ IES (literal) on CIFAR-10, $2.10\times$ on CIFAR-100, tight seed variance on savings specifically. Self-calibration, not the fixed- δ methods’ hand-tuned constant, is the mechanism: δ rescales $186\times$ (CIFAR-10) and $333\times$ (CIFAR-100) with no constant retuned by hand. Absolute savings are much lower than CNNs ($\sim 26\%$ vs. $\sim 54.5\%$ DenseNet-121). There is a small accuracy cost on transformers unlike CNNs (0.36–0.69 points, widest seed spread of any method), suggesting instance removal destabilizes a data-starved, from-scratch ViT more than a CNN.

Caveats. K^* was pinned at the configured ceiling (15) in every ViT run: Corollary 7.3’s closed form asked for more patience than allowed, so these numbers reflect the theory clamped by configuration; lifting the cap is untested. Savings arrive late (exclusions ramp steeply only in the second half of training). δ recalibrates once, at epoch 30 as in every CNN cell; whether this timing is also right for a transformer’s different loss dynamics is untested.

Absolute accuracy is well below the CNN cells, expected for a from-scratch transformer on 50k images with crop-and-flip augmentation only.

R K^* -Sweep: Full Trajectory Figure and Discussion

Table 10: K^* -sweep, primary cell, mean $\pm$ std, 3 seeds, δ self-calibrated. Matched baseline: 95.08 $\pm$ 0.14% acc.

K^* (fixed)	Acc. (%)	Saved (%)
1	94.55 $\pm$ 0.16	66.98 $\pm$ 0.62
2	94.56 $\pm$ 0.08	65.75 $\pm$ 0.27
4	94.62 $\pm$ 0.10	63.24 $\pm$ 0.44
8	94.80 $\pm$ 0.09	58.85 $\pm$ 0.57
15	94.80 $\pm$ 0.09	53.38 $\pm$ 0.58
30	94.87 $\pm$ 0.09	44.13 $\pm$ 0.40

Trajectory divergence from the shared full-data baseline reference, primary cell, 3-seed mean

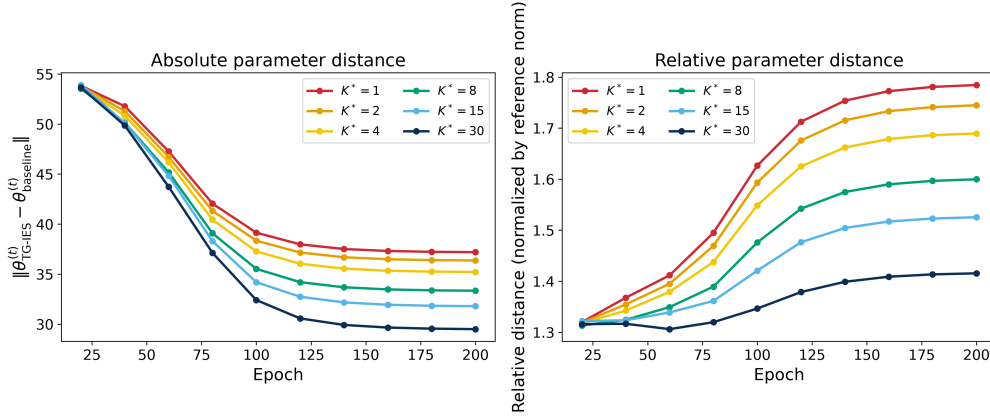


Figure 11: Parameter-space distance between TG-IES’s actual trajectory and the shared-seed baseline reference vs. epoch, one curve per fixed K^* , 3-seed mean. Left: absolute distance. Right: normalized by the baseline’s own parameter norm at that epoch.

Absolute parameter distance from baseline decreases over training for every K^* tested, from ≈ 54 at epoch 20 to 29–37 by epoch 200, plateauing over the last third of training. At every snapshot epoch the ordering across K^* matches Theorem 7.1’s predicted direction exactly, with no crossings. The normalized distance (right panel) instead rises and plateaus higher, because the reference parameter norm itself shrinks faster (weight decay) than the absolute gap does; we report both panels since which normalization is the “right” one to read Remark 7.2’s open question through is not settled by this experiment. This is empirical evidence consistent with, but not a substitute for, a genuine multi-step bound (Appendix D); we update Section 12’s framing from “untested” to “empirically consistent with, formally still open.”

S Naive Grid Search Over δ : Is the Calibration Better Than Search?

Section 10.1’s “ δ only” row shows calibrating δ alone (Corollary 5.3, $K^*=1$) reaches 94.68% acc. / 66.62% saved. This does not by itself show the calibration is doing useful work rather than TG-IES simply landing, by whatever means, on a more aggressive threshold than IES’s hand-tuned 0.001 – a question a theory-guided calibration paper should not leave unanswered. We test it directly: fixing $K^*=1$ and sweeping δ over a log-spaced grid (primary cell, seed 0, matching the ablation’s own single-seed protocol), with no calibration involved at all.

Finding: the calibrated value sits on the same frontier a naive grid traces, not above it. Accuracy falls and savings rise smoothly and roughly monotonically across the whole grid (Table 11), the ordinary accuracy–compute trade-off, not a special curve the calibration alone accesses. The calibrated point does not Pareto-dominate every grid point (at similar savings, $\delta=0.015$ reaches 70.47% saved at 94.52% acc., a different but comparable trade-off

Table 11: Naive δ grid search, $K^*=1$ fixed, primary cell, seed 0. The calibrated row (from Table 5) is included for comparison, not part of the grid.

δ	Acc. (%)	Saved (%)
0.001 (IES’s own constant)	94.92	48.59
0.003	94.76	61.38
0.005	94.39	64.75
0.0077 (calibrated, Corollary 5.3)	94.68	66.62
0.015	94.52	70.47
0.03	94.27	72.98
0.05	93.92	74.48

to the calibrated point’s 66.62%/94.68%) nor is it dominated by every grid point (it beats $\delta=0.005$ on both axes). Reading this honestly: this experiment does not demonstrate that Corollary 5.3’s derivation finds a better δ than an untheorized search would, at comparable final quality. What it does demonstrate is a cheaper route to a comparable point: the calibration reaches this frontier in a single run (via one cheap recalibration pass at epoch 30), where the naive grid above needed five full 200-epoch runs to trace the same region. We had not run this comparison before a reviewer asked for it, and we do not consider the question closed by one seed, one architecture, and a five-point grid – a denser sweep, repeated across seeds, and on a setting where TG-IES’s calibration is known to underperform (e.g. Appendix J) would be needed to check whether this conclusion (comparable quality, cheaper to reach) holds more broadly, or whether the calibration’s real advantage is narrower than that.

T Reactivation Probe: Does the Excluded Set Actually Satisfy the Gradient-Floor Hypothesis?

Theorem 7.1’s bound and Algorithm 1’s derivation of K^* both rest on a specific per-instance hypothesis: every currently-excluded instance i satisfies $\|g_i^{(t)}\| < \varepsilon_0$, where ε_0 is the cheap, last-layer proxy percentile estimated once at epoch 30 (Appendix H). Every check of this hypothesis so far in this paper (Section 7.1’s empirical check, Appendix P) has used the same cheap proxy or a correlational check, not the true full-model gradient norm on the actual excluded set during an otherwise-unmodified run. We close that gap directly: --reactivation_probe_every 20 adds, every 20 epochs, an individual batch-size-1 forward+backward pass on a random subsample of 300 currently-passive instances (self-calibrated δ , primary cell, 3 seeds, tagged _reactprobe), computing each one’s true $\|g_i^{(t)}\|$ and checking it against the run’s own calibrated ε_0 . This instrumentation is purely diagnostic: it does not feed back into the algorithm’s actual removal or reactivation decisions, so the probed runs’ accuracy and savings ($94.67 \pm 0.11\%$ acc., $53.66 \pm 0.10\%$ saved) match the unprobed primary-cell TG-IES numbers (Table 3) closely, confirming the probe itself does not perturb training.

Table 12: Reactivation probe, primary cell, 3-seed mean of frac_exceed_eps0 (fraction of a 300-instance random subsample of the passive set whose true gradient norm exceeds the run’s calibrated ε_0) at each probed epoch.

Epoch	40	60	80	100	120	140	160	180	200
Frac. $> \varepsilon_0$	0.52	0.48	0.39	0.46	0.48	0.46	0.47	0.50	0.50

Finding: roughly half the excluded set exceeds the calibrated floor at every check, with no decreasing trend. Pooling all 9 probed epochs across 3 seeds (27 (epoch, seed) points, 300 probed instances each), the mean fraction of the passive set exceeding ε_0 is 47.4%, ranging 34–59% epoch-to-epoch with no consistent decrease over training (Table 12) – if anything, the fraction is lowest at epoch 80 and drifts back up to $\approx 50\%$ by epoch 200, the opposite of what a hypothesis that becomes more valid as training progresses would predict. This directly complicates Theorem 7.1’s per-instance hypothesis in practice: at any given check, roughly half of the instances the algorithm currently treats as safely excluded have a true gradient norm the theory’s own floor says they should not. The picture is not uniformly bad, however: each seed’s median probed gradient norm (0.025–0.029 across seeds and epochs) sits close to that seed’s own ε_0 (0.027–0.030), while the mean probed gradient norm is 2–

10 $\times$ higher (0.07–0.29) – a heavy right tail, not a uniformly shifted distribution. So the calibrated ε_0 is a reasonable estimate of a typical (median) excluded instance’s gradient floor, but a systematically loose one for the population as a whole, since roughly half the excluded set sits above it at any moment.

Reading this alongside Section 10.2. Section 10.2’s aggregate trajectory-divergence measurement, on the same primary cell, finds absolute parameter drift from the full-data baseline plateaus rather than growing without bound, despite this per-instance violation rate never approaching zero. We do not have an explanation for this reconciliation that we are confident in and do not offer one we have not verified: a natural candidate is that Theorem 7.1’s bound is itself an average, $1/N$ -scaled quantity, so a near-half violation rate at a bounded excess magnitude (the heavy tail above, not an unbounded one) may still keep the aggregate update small relative to $N = 50,000$. Appendix E now derives and checks this quantitatively rather than leaving it as an untested candidate. What this probe does establish is narrower and more direct: the gradient-floor hypothesis behind Theorem 7.1 and Corollary 7.3 is a population-level approximation that a large fraction of individual excluded instances violate at any snapshot, not a hypothesis that holds instance-by-instance as the theorem’s statement (“every currently-excluded instance satisfies $\|g_i^{(t)}\| < \varepsilon$ ”) would suggest if read literally. We add this to Section 12 as a further qualification of the bound’s practical scope, alongside the already-open multi-step question (Remark 7.2), rather than treat Section 10.2’s aggregate result as having closed it.

U Extended Future-Work List

(1) Scale. All experiments are CIFAR-scale; TinyImageNet or ImageNet-100 would test whether calibration (in particular the gradient-norm-floor proxy behind K^*) transfers to higher resolution and more classes. (2) Architecture family, partly closed. Appendix Q adds one CIFAR-native ViT-Tiny; a larger transformer (ViT-Small, DeiT-Tiny) or one at a resolution where patch-16 tokenization is meaningful remains untried, as does raising `--max_kstar`. (3) Optimizer/schedule sensitivity. Section 9 and Appendix J show TG-IES trailing simpler IES variants on settings without the exponential schedule; Appendix K’s candidate mechanism is only partly confirmed by direct instrumentation, and a controlled ablation calibrating δ, K^* under a matched noise/gradient-norm profile across schedules remains unrun. The bridge’s qualitative direction ($\|g_i^{(t)}\|$ vs. $\sqrt{|\Delta^2 L_i^{(t)}|}$) is now confirmed across all three schedules; only the exact bound $\varepsilon_i(\delta)$, needing per-instance r_i, Λ_i , remains open. (4) RHO-LOSS (Mindermann et al., 2022) and RL-style dynamic pruning (Zhu et al., 2026) are not compared. (5) The multi-step trajectory-divergence bound, now empirically probed (Appendix R), formally attempted (Appendix D, exponentially loose), and partly refined at the aggregate level (Appendix E: a provably tighter, directly-measured leading constant, still the same exponential structural form) but not proved tightly – we view this as the single most important remaining theoretical gap, ahead of (1)–(4).

V Additional Experimental Details

Full per-run results. Table 13 lists every IES-family run underlying the main tables and Appendices J and Q (96 runs total). Two rows (ResNet-18/F and VGG-16/E, TG-IES, seed 0) supersede values from an earlier version traced to the reproducibility issue of Section 9; stale originals are retained in the released code’s results/ directory with a `_STALE_preverif` suffix for audit purposes.

Table 13: All 96 IES-family runs. E = SGD exponential LR, F = SGD fixed LR, C100 = CIFAR-100, ViT/C10 and ViT/C100 = ViT-Tiny on CIFAR-10/CIFAR-100. Color on TG-IES rows compares against the other methods for that same seed.

Setting	Method	Seed	Acc. (%)	Saved (%)
ResNet-18/E	Baseline	0	94.96	0.03
ResNet-18/E	Baseline	1	95.18	0.03
ResNet-18/E	Baseline	2	94.95	0.03

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Table 13 – continued from previous page

Setting	Method	Seed	Acc. (%)	Saved (%)
ResNet-18/E	IES (shipped)	0	94.61	45.36
ResNet-18/E	IES (shipped)	1	94.74	46.09
ResNet-18/E	IES (shipped)	2	94.79	45.53
ResNet-18/E	IES (literal)	0	94.92	48.59
ResNet-18/E	IES (literal)	1	94.66	49.00
ResNet-18/E	IES (literal)	2	94.97	49.47
ResNet-18/E	TG-IES	0	94.61	53.48
ResNet-18/E	TG-IES	1	95.16	53.44
ResNet-18/E	TG-IES	2	94.77	53.98
ResNet-18/F	Baseline	0	92.28	0.03
ResNet-18/F	Baseline	1	92.21	0.03
ResNet-18/F	Baseline	2	92.13	0.03
ResNet-18/F	IES (shipped)	0	91.78	44.71
ResNet-18/F	IES (shipped)	1	91.69	44.55
ResNet-18/F	IES (shipped)	2	91.61	44.23
ResNet-18/F	IES (literal)	0	91.77	49.60
ResNet-18/F	IES (literal)	1	91.66	49.38
ResNet-18/F	IES (literal)	2	91.87	49.27
ResNet-18/F	TG-IES	0	91.71	29.57
ResNet-18/F	TG-IES	1	91.96	29.27
ResNet-18/F	TG-IES	2	92.07	28.98
VGG-16/E	Baseline	0	93.25	0.03
VGG-16/E	Baseline	1	93.21	0.03
VGG-16/E	Baseline	2	93.44	0.03
VGG-16/E	IES (shipped)	0	93.33	38.67
VGG-16/E	IES (shipped)	1	93.27	37.46
VGG-16/E	IES (shipped)	2	93.34	36.99
VGG-16/E	IES (literal)	0	93.04	41.85
VGG-16/E	IES (literal)	1	93.35	41.45
VGG-16/E	IES (literal)	2	93.07	39.74
VGG-16/E	TG-IES	0	92.96	54.36
VGG-16/E	TG-IES	1	93.15	54.56
VGG-16/E	TG-IES	2	93.23	54.37
DenseNet-121/E	Baseline	0	95.37	0.03
DenseNet-121/E	Baseline	1	95.50	0.03
DenseNet-121/E	Baseline	2	95.22	0.03
DenseNet-121/E	IES (shipped)	0	95.37	48.19
DenseNet-121/E	IES (shipped)	1	95.47	49.79
DenseNet-121/E	IES (shipped)	2	95.25	47.83
DenseNet-121/E	IES (literal)	0	95.26	51.96
DenseNet-121/E	IES (literal)	1	95.24	52.27
DenseNet-121/E	IES (literal)	2	95.23	50.59
DenseNet-121/E	TG-IES	0	95.24	54.73
DenseNet-121/E	TG-IES	1	95.22	55.06
DenseNet-121/E	TG-IES	2	95.38	53.79
ResNet-18/Adam	Baseline	0	92.98	0.03
ResNet-18/Adam	Baseline	1	92.88	0.03
ResNet-18/Adam	Baseline	2	92.86	0.03
ResNet-18/Adam	IES (shipped)	0	92.67	56.04
ResNet-18/Adam	IES (shipped)	1	92.50	55.80
ResNet-18/Adam	IES (shipped)	2	92.66	56.23
ResNet-18/Adam	IES (literal)	0	92.49	59.80
ResNet-18/Adam	IES (literal)	1	92.47	60.39
ResNet-18/Adam	IES (literal)	2	92.30	60.10
ResNet-18/Adam	TG-IES	0	92.33	30.86
ResNet-18/Adam	TG-IES	1	92.49	29.64
ResNet-18/Adam	TG-IES	2	92.51	30.04
ResNet-18/C100	Baseline	0	76.46	0.03
ResNet-18/C100	Baseline	1	77.27	0.03
ResNet-18/C100	Baseline	2	76.91	0.03
ResNet-18/C100	IES (shipped)	0	76.64	11.74
ResNet-18/C100	IES (shipped)	1	76.69	11.99
ResNet-18/C100	IES (shipped)	2	77.29	12.19
ResNet-18/C100	IES (literal)	0	76.33	18.18
ResNet-18/C100	IES (literal)	1	76.92	18.32

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Setting	Method	Seed	Acc. (%)	Saved (%)
ResNet-18/C100	IES (literal)	2	77.09	18.98
ResNet-18/C100	TG-IES	0	75.97	57.81
ResNet-18/C100	TG-IES	1	75.96	57.73
ResNet-18/C100	TG-IES	2	76.11	57.75
ViT-Tiny/C10	Baseline	0	84.25	0.03
ViT-Tiny/C10	Baseline	1	84.30	0.03
ViT-Tiny/C10	Baseline	2	83.67	0.03
ViT-Tiny/C10	IES (shipped)	0	84.13	13.37
ViT-Tiny/C10	IES (shipped)	1	83.83	12.97
ViT-Tiny/C10	IES (shipped)	2	83.81	13.06
ViT-Tiny/C10	IES (literal)	0	84.32	17.05
ViT-Tiny/C10	IES (literal)	1	83.90	16.48
ViT-Tiny/C10	IES (literal)	2	83.37	16.52
ViT-Tiny/C10	TG-IES	0	84.26	25.80
ViT-Tiny/C10	TG-IES	1	83.47	25.30
ViT-Tiny/C10	TG-IES	2	83.41	25.45
ViT-Tiny/C100	Baseline	0	59.19	0.03
ViT-Tiny/C100	Baseline	1	59.51	0.03
ViT-Tiny/C100	Baseline	2	59.05	0.03
ViT-Tiny/C100	IES (shipped)	0	58.96	8.20
ViT-Tiny/C100	IES (shipped)	1	59.40	8.62
ViT-Tiny/C100	IES (shipped)	2	59.20	8.33
ViT-Tiny/C100	IES (literal)	0	58.54	12.28
ViT-Tiny/C100	IES (literal)	1	59.02	12.70
ViT-Tiny/C100	IES (literal)	2	58.75	12.55
ViT-Tiny/C100	TG-IES	0	58.97	26.23
ViT-Tiny/C100	TG-IES	1	58.05	26.43
ViT-Tiny/C100	TG-IES	2	58.66	26.17

Hardware and reproducibility. All runs used a single NVIDIA RTX 5090 GPU; Appendix O reports precise wall-clock, memory, utilization, and estimated energy. All four methods are implemented in a single training entry point taking `--method {baseline,ies_shipped,ies_alg1,tgies}` so data pipeline, model, and optimizer code are shared exactly across methods; a single script drives the full experimental matrix and supports resuming any interrupted run.

W Statistical Analysis of Seed-Level Comparisons

Every $\pm$ figure elsewhere in this paper is a sample standard deviation over 3 seeds, not a significance test. This section reports one directly. `setup_seed(seed)` is called once per run, before model initialization and data loading, so all four methods sharing a seed share the same initial weights and data-shuffling order – a genuine paired design, analyzed via paired per-seed differences rather than treating (method, seed) runs as independent. Eight settings have 3 seeds: the primary cell, both secondary CNN cells, Adam, CIFAR-100, DenseNet-121, and both ViT-Tiny cells; the three LR-sweep cells of Appendix K remain single-seed by design.

For each 3-seed setting we compare TG-IES against each other method on best test accuracy, backprop-saved %, and FLOP-saved %, reporting: the paired mean difference and its sample standard deviation; a 95% CI via the exact t -distribution at 2 degrees of freedom ($t_{0.975,2} = 4.303$); Cohen’s d_z (paired effect size); and an exact two-sided sign test (floor $p = 2/8 = 0.25$ at $n = 3$, reached whenever all seeds agree in sign, regardless of effect size).

Table 14: Paired per-seed statistics, TG-IES vs. each other method, every 3-seed setting ($n = 3$). CI: 95%, $t(df=2)$. d_z : Cohen’s paired effect size. p_{sign} : exact two-sided sign test (floor 0.25 at $n = 3$).

Metric	Comparison (TG-IES –)	Mean Δ	95% CI	d_z	p_{sign}
Primary: ResNet-18, CIFAR-10, SGD (exponential LR)					

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Metric	Comparison (TG-IES –)	Mean Δ	95% CI	d_z	p_{sign}
Acc. (%)	Baseline	–0.18	[–0.59, 0.23]	–1.11	0.250
Acc. (%)	IES (shipped)	0.13	[–0.48, 0.75]	0.54	1.000
Acc. (%)	IES (literal)	0.00	[–1.09, 1.09]	–0.01	1.000
Backprop Saved (%)	Baseline	53.60	[52.86, 54.34]	179.2	0.250
Backprop Saved (%)	IES (shipped)	7.97	[6.57, 9.37]	14.2	0.250
Backprop Saved (%)	IES (literal)	4.61	[4.00, 5.22]	18.8	0.250
FLOP Saved (%)	Baseline	35.71	[35.22, 36.21]	179.2	0.250
FLOP Saved (%)	IES (shipped)	5.31	[4.38, 6.24]	14.2	0.250
FLOP Saved (%)	IES (literal)	3.07	[2.67, 3.48]	18.8	0.250
Secondary: ResNet-18, CIFAR-10, SGD (fixed LR)					
Acc. (%)	Baseline	–0.29	[–0.93, 0.35]	–1.14	0.250
Acc. (%)	IES (shipped)	0.22	[–0.45, 0.89]	0.82	1.000
Acc. (%)	IES (literal)	0.15	[–0.31, 0.61]	0.79	1.000
Backprop Saved (%)	Baseline	29.24	[28.51, 29.98]	98.8	0.250
Backprop Saved (%)	IES (shipped)	–15.23	[–15.42, –15.03]	–194.1	0.250
Backprop Saved (%)	IES (literal)	–20.14	[–20.48, –19.80]	–148.1	0.250
FLOP Saved (%)	Baseline	19.49	[18.99, 19.97]	98.8	0.250
FLOP Saved (%)	IES (shipped)	–10.14	[–10.27, –10.02]	–194.1	0.250
FLOP Saved (%)	IES (literal)	–13.42	[–13.65, –13.20]	–148.1	0.250
Secondary: VGG-16, CIFAR-10, SGD (exponential LR)					
Acc. (%)	Baseline	–0.19	[–0.48, 0.10]	–1.60	0.250
Acc. (%)	IES (shipped)	–0.20	[–0.57, 0.17]	–1.36	0.250
Acc. (%)	IES (literal)	–0.04	[–0.50, 0.42]	–0.22	1.000
Backprop Saved (%)	Baseline	54.40	[54.11, 54.69]	470.5	0.250
Backprop Saved (%)	IES (shipped)	16.72	[14.46, 18.98]	18.4	0.250
Backprop Saved (%)	IES (literal)	13.42	[10.69, 16.13]	12.3	0.250
FLOP Saved (%)	Baseline	36.23	[36.04, 36.43]	470.5	0.250
FLOP Saved (%)	IES (shipped)	11.14	[9.63, 12.64]	18.4	0.250
FLOP Saved (%)	IES (literal)	8.94	[7.12, 10.75]	12.3	0.250
Adam: ResNet-18, CIFAR-10					
Acc. (%)	Baseline	–0.46	[–0.87, –0.06]	–2.84	0.250
Acc. (%)	IES (shipped)	–0.17	[–0.58, 0.24]	–1.01	0.250
Acc. (%)	IES (literal)	0.02	[–0.44, 0.48]	0.13	1.000
Backprop Saved (%)	Baseline	30.15	[28.61, 31.69]	48.6	0.250
Backprop Saved (%)	IES (shipped)	–25.85	[–27.28, –24.41]	–44.7	0.250
Backprop Saved (%)	IES (literal)	–29.91	[–32.18, –27.65]	–32.8	0.250
FLOP Saved (%)	Baseline	20.09	[19.06, 21.11]	48.6	0.250
FLOP Saved (%)	IES (shipped)	–17.22	[–18.18, –16.26]	–44.7	0.250
FLOP Saved (%)	IES (literal)	–19.93	[–21.44, –18.42]	–32.8	0.250
CIFAR-100: ResNet-18, SGD (exponential LR)					
Acc. (%)	Baseline	–0.87	[–1.90, 0.16]	–2.09	0.250
Acc. (%)	IES (shipped)	–0.86	[–1.55, –0.17]	–3.09	0.250
Acc. (%)	IES (literal)	–0.77	[–1.64, 0.11]	–2.18	0.250
Backprop Saved (%)	Baseline	57.73	[57.64, 57.83]	1522.5	0.250
Backprop Saved (%)	IES (shipped)	45.79	[45.15, 46.43]	178.2	0.250
Backprop Saved (%)	IES (literal)	39.27	[38.16, 40.37]	88.3	0.250
FLOP Saved (%)	Baseline	38.47	[38.40, 38.53]	1522.5	0.250
FLOP Saved (%)	IES (shipped)	30.51	[30.08, 30.94]	178.2	0.250
FLOP Saved (%)	IES (literal)	26.17	[25.43, 26.90]	88.3	0.250
DenseNet-121: CIFAR-10, SGD (exponential LR)					
Acc. (%)	Baseline	–0.08	[–0.64, 0.47]	–0.37	1.000
Acc. (%)	IES (shipped)	–0.08	[–0.57, 0.40]	–0.43	1.000
Acc. (%)	IES (literal)	0.04	[–0.21, 0.28]	0.37	1.000
Backprop Saved (%)	Baseline	54.49	[52.85, 56.13]	82.5	0.250
Backprop Saved (%)	IES (shipped)	5.92	[4.36, 7.49]	9.4	0.250
Backprop Saved (%)	IES (literal)	2.92	[2.32, 3.53]	12.0	0.250
FLOP Saved (%)	Baseline	36.32	[35.22, 37.41]	82.5	0.250
FLOP Saved (%)	IES (shipped)	3.95	[2.90, 4.99]	9.4	0.250
FLOP Saved (%)	IES (literal)	1.95	[1.54, 2.35]	12.0	0.250

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Metric	Comparison (TG-IES –)	Mean Δ	95% CI	d_z	p_{sign}
ViT-Tiny: CIFAR-10, AdamW (warmup-cosine)					
Acc. (%)	Baseline	−0.36	[−1.43, 0.71]	−0.84	1.000
Acc. (%)	IES (shipped)	−0.21	[−0.94, 0.52]	−0.71	1.000
Acc. (%)	IES (literal)	−0.15	[−0.77, 0.47]	−0.61	1.000
Backprop Saved (%)	Baseline	25.49	[24.84, 26.13]	98.5	0.250
Backprop Saved (%)	IES (shipped)	12.38	[12.25, 12.52]	228.0	0.250
Backprop Saved (%)	IES (literal)	8.83	[8.62, 9.05]	101.1	0.250
FLOP Saved (%)	Baseline	17.06	[16.63, 17.49]	98.5	0.250
FLOP Saved (%)	IES (shipped)	8.29	[8.20, 8.38]	228.0	0.250
FLOP Saved (%)	IES (literal)	5.91	[5.77, 6.06]	101.1	0.250
ViT-Tiny: CIFAR-100, AdamW (warmup-cosine)					
Acc. (%)	Baseline	−0.69	[−2.36, 0.98]	−1.03	0.250
Acc. (%)	IES (shipped)	−0.63	[−2.33, 1.07]	−0.92	1.000
Acc. (%)	IES (literal)	−0.21	[−1.97, 1.55]	−0.30	1.000
Backprop Saved (%)	Baseline	26.25	[25.90, 26.59]	191.0	0.250
Backprop Saved (%)	IES (shipped)	17.89	[17.60, 18.18]	154.5	0.250
Backprop Saved (%)	IES (literal)	13.77	[13.36, 14.17]	85.0	0.250
FLOP Saved (%)	Baseline	17.57	[17.34, 17.79]	191.0	0.250
FLOP Saved (%)	IES (shipped)	11.98	[11.78, 12.17]	154.5	0.250
FLOP Saved (%)	IES (literal)	9.21	[8.94, 9.48]	85.0	0.250

What this does and does not support. On the primary cell, TG-IES’s backprop- and FLOP-saved advantage over every other method is in the same direction in all 3 seeds for every comparison (hence $p_{\text{sign}} = 0.25$, the floor), with CIs entirely on one side of zero; large d_z values (12–470) reflect small seed-to-seed variability relative to a large, consistent savings effect. The same pattern holds on VGG-16, CIFAR-100 ($d_z = 1522.5$, the largest in this table), DenseNet-121, and both ViT-Tiny cells. On ResNet-18(fixed-LR) and Adam, the sign flips as Section 9 and Appendix J report qualitatively, with CIs again excluding zero entirely. Accuracy differences are small ($|\text{mean } \Delta| \leq 0.87$ points) with CIs mostly straddling zero; the exceptions (Adam vs. Baseline, CIFAR-100 vs. IES shipped) are real, if small, effects at this sample size. A CI containing zero is not evidence the true difference is zero, and a large d_z with $p_{\text{sign}} = 0.25$ is not conventionally “significant” – it is the most a 3-seed exact sign test can report regardless of effect size; we do not describe any result here as statistically significant at a threshold this design cannot reach.